

ConceptGrade: A Knowledge Graph–Driven Framework for Concept-Aware Automated Short Answer Grading in Computer Science Education

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Abstract—Correction notice (2026-07-28). An earlier version of this paper reported Mohler-dataset results computed against a hand-authored, fabricated 120-sample fixture rather than the real Mohler et al. (2011) benchmark; the discovery, verification, and full incident record are in REPRODUCIBILITY.md. This Abstract and Introduction now report the corrected, real-data Mohler result (46 questions, 1,262 responses, sourced from the public `nkazi/MohlerASAG` mirror, CC-BY-4.0) and the cross-dataset meta-analysis recomputed with it in place of the fabricated numbers. The original six-condition component ablation required new pipeline runs on real data and is reported as open future work; however, a real-data 3-condition ablation (C_LLM, pre-verifier `kg_score`, C5_fix) has since been reconstructed from already-cached components at zero new API cost (§VI) and found the KG-grounded score, taken alone, to be substantially *worse* than the zero-shot baseline — essentially all of ConceptGrade’s small real-data gain comes from the verifier stage, not the KG comparison. The frozen Sentence-BERT baselines have been re-verified on real data (§V-A).

This paper is an empirical, cautionary study of the fragility and domain boundaries of hybrid Knowledge-Graph + Large Language Model (KG-LLM) pipelines for Automated Short Answer Grading (ASAG) in Computer Science education. We present ConceptGrade, a five-layer concept-aware ASAG framework that transforms student responses into *concept graphs* and compares them against expert-curated domain knowledge graphs (KGs). The system extracts concepts and relationships from free-text answers using a chain-of-thought LLM pipeline, computes semantic coverage and structural integration scores via graph matching, and synthesizes a composite grade through a weighted scoring model. We construct an expert Data Structures KG comprising 101 domain concepts and 138 typed relationships spanning eight semantic categories. We evaluate ConceptGrade on $n = 2,276$ graded student answers spanning 213 questions across three independent ASAG datasets: Mohler 2011 (CS Data Structures, $n = 1,262$, 46 questions, real data as corrected above), DigiKlausur (Neural Networks, $n = 646$, 17 questions), and Kaggle ASAG (Elementary Science, $n = 368$ unique responses, 150 questions — the source distribution’s 473 records contain 105 byte-identical duplicates that inflate variance if kept). ConceptGrade achieves a modest, statistically fragile in-domain improvement over the identical-model LLM baseline on the real Mohler data (8.2% MAE reduction, significant at the response level but only marginal at the question-clustered level), and cross-dataset evaluation across DigiKlausur and Kaggle ASAG confirms the

effect does not robustly generalise. On the real, KG-aligned Mohler sample ($n = 1,262$; every response is effectively held out, since the pipeline’s hyperparameters were fixed before this real dataset was used), ConceptGrade achieves Pearson $r = 0.784$ (LLM baseline: 0.790 — ConceptGrade’s correlation is *worse*), QWK = 0.524 (baseline 0.501), and MAE = 1.177 (baseline 1.282), an 8.2% MAE reduction (Wilcoxon two-tailed $p < 0.0001$; paired $d_z = -0.154$, a small effect). This response-level significance should not be over-read: at the question-clustered level (46 real questions, the more appropriate unit given non-independence within a question), the effect is only marginal (two-tailed $p = 0.111$, one-tailed $p = 0.056$), and ConceptGrade has lower mean error on just 27 of 46 questions (58.7%); only 17 of 46 leave-one-question-out folds reach one-tailed significance. Both systems also systematically under-predict relative to real human graders (mean human score 4.24/5 vs. 2.99 for the LLM baseline and 3.09 for ConceptGrade), indicating real course grading is markedly more lenient than either LLM-based grader. A post-hoc, 5-fold cross-validated monotonic recalibration (§V-B) fit only on already-collected scores (zero new LLM calls) cuts MAE by roughly $3\times$ for both systems ($1.282 \rightarrow 0.375$ for C_LLM, $1.177 \rightarrow 0.392$ for ConceptGrade) — most of the raw error was fixable scale bias, not ranking error — and *inverts* the comparison: once both are properly calibrated, C_LLM has significantly *lower* MAE than ConceptGrade ($p = 0.017$ one-tailed), consistent with ConceptGrade’s raw MAE edge being substantially a calibration artifact rather than evidence of better grading judgment. Pooling across all three datasets (fixed-effects inverse-variance meta-analysis, $n = 2,276$, 213 questions): ConceptGrade improves over the baseline on Mohler ($d_z = -0.154$, $p_{\text{two}} < 0.0001$), shows a small but significant improvement on DigiKlausur ($d_z = -0.067$, $p_{\text{one}} = 0.024$, practical effect below Cohen’s small-effect threshold), and shows no significant improvement on Kaggle ASAG ($d_z = -0.007$, $p_{\text{one}} = 0.351$, $p_{\text{two}} = 0.702$ on $N = 368$ unique responses after removing 105 byte-identical duplicate records from the source distribution). The pooled fixed-effects estimate is $d_z = -0.105$ (95% CI $[-0.147, -0.064]$, $p_{\text{two}} < 0.0001$), but between-dataset heterogeneity is substantial ($I^2 = 73.2\%$, Higgins–Thompson sense) and the methodologically more appropriate random-effects estimate is $d_z = -0.084$ with a 95% CI $[-0.169, +0.002]$ that essentially touches zero — we do not claim the pooled effect as confirmatory evidence of generalisation. On Kaggle ASAG the LLM extraction module produced empty matched-concept sets for 368/368 unique answers (473/473 on the raw, pre-deduplication

pipeline run), an upstream signal correctly identified by the pipeline as OUT-OF-KG-DOMAIN and now explicitly surfaced to downstream consumers rather than silently defaulting to vacuous-perfect coverage scores; the KG-grounding effect requires a non-empty KG-answer intersection, and elementary-science colloquial language does not provide one. We treat the three-dataset behaviour as a *boundary characterisation*, not a generalisation claim. The six-condition component ablation and non-LLM neural baselines reported in an earlier draft were computed on the fabricated Mohler fixture and require new pipeline runs to re-verify on real data; we report them as open future work rather than carry forward numbers that may not hold. Self-consistency ensembling (§V-F: issuing 7 independent gradings at temperature 0.7 and mean-aggregating, holding all other inputs fixed; a design choice made *before* touching the evaluation data, unlike an earlier linear-blend alternative that failed cross-validation and is reported as retracted, not as a finding) improves response-level accuracy on both winning datasets but does not establish question-level robustness once compared against an equally-resourced baseline, checked on both. Against a single-call C_LLM, every leave-one-question-out fold reaches significance at both tails on Mohler (50/50) and DigiKlausur (17/17); a fair-control check on *both* datasets — giving C_LLM the identical $7\times$ /temperature-0.7/mean-aggregation treatment rather than comparing against a single call — shows the MAE gap persists on both (+7.0% Mohler, +6.4% DigiKlausur, both $p < 0.001$), but question-clustered significance and LOOCV robustness largely do not: Mohler collapses to $p = 0.256$ and 0/46 folds; DigiKlausur falls to $p = 0.089$ two-tailed (0.044 one-tailed) and 5/17/2/17 folds. Pearson correlation, worse than baseline for the single-call system, exceeds a single-call baseline on both winning datasets (0.790 vs. 0.782; 0.727 vs. 0.701); against the fair, equally-resourced control this advantage narrowly survives on Mohler (0.797 vs. 0.790) but *reverses* on DigiKlausur (0.727 vs. 0.735). Neither self-consistency variant extends to Kaggle ASAG, consistent with that dataset’s already-established lack of KG-grounded signal to denoise — a domain-boundary result, not a failure of the method. This is, at real cost ($7\times$ inference calls) and with caveats that persist on both datasets (Spearman rank correlation trails baseline), a real but substantially more modest positive result than initially reported: a precisely-estimated response-level improvement over a fairly-resourced baseline on two datasets, with question-level robustness and (on DigiKlausur) correlation superiority established only against a single-call baseline. These results together demonstrate that KG-aware grading provides, at best, a small gain over the identical-model LLM baseline in its favourable vocabulary-specific domains — fragile with a single call, and only partially more robust with self-consistency ensembling once the comparison is made fair — and characterise the boundary conditions beyond which the effect disappears entirely.

Index Terms—automated short answer grading, knowledge graph, concept extraction, computer science education, large language models, Bloom’s taxonomy, NLP

I. INTRODUCTION

Scope and framing (this is a fragility-and-boundaries study, not a state-of-the-art claim). This paper does not propose that multi-layer knowledge-graph grounding is a generalised solution for short-answer grading. Our empirical evidence shows the opposite: the multi-layer architecture yields, at best, a small and statistically fragile improvement in its single most favourable in-domain regime, and is neutral or negatively-contributing in adjacent-domain settings. Our claims are therefore confined to a pooled $n = 2,276$ paired predictions

across three independent ASAG datasets and 213 questions (Section V-D): (i) an 8.2% paired-MAE reduction on the real, KG-aligned Mohler 2011 sample ($n = 1,262$, 46 questions, sourced from the public `nkazi/MohlerASAG` mirror after an earlier fabricated-data error was discovered and corrected — see `REPRODUCIBILITY.md`) against the identical zero-shot LLM ($d_z = -0.154$, two-tailed $p < 0.0001$ at the response level), which weakens to only marginal significance at the question-clustered level (46 questions, two-tailed $p = 0.111$, one-tailed $p = 0.056$; only 17/46 leave-one-question-out folds reach one-tailed significance, and ConceptGrade has lower mean error on just 27/46 questions); Pearson correlation is in fact *worse* for ConceptGrade ($r = 0.784$) than for the baseline ($r = 0.790$); (ii) a real-data 3-condition ablation (§VI), reconstructed from already-cached components at zero new API cost, finds the KG-grounded pre-verifier score substantially *worse* than the zero-shot baseline (−86.9% MAE) and attributes essentially all of ConceptGrade’s small net gain to the verifier stage rather than KG grounding; the original fabricated-data six-condition ablation is retracted and the non-LLM neural baselines have been re-verified on real data (§V-A); (iii) a fixed-effects cross-dataset pooled effect ($d_z = -0.105$, $p_{\text{two}} < 0.0001$) with substantial between-dataset heterogeneity ($I^2 = 73.2\%$) whose random-effects 95% CI $[-0.169, +0.002]$ essentially touches zero — the multi-layer pipeline does not robustly generalise; (iv) architectural failure-mode evidence: the upstream LLM concept-extraction module produced empty matched-concept sets on 368/368 unique Kaggle ASAG samples (473/473 on the raw, pre-deduplication pipeline run), predicting the null effect on that dataset directly from the KG-vocabulary mismatch; (v) self-consistency ensembling (§V-F: 7 independent gradings at temperature 0.7, mean-aggregated, design fixed before touching the evaluation data) improves response-level MAE over a single-call baseline on Mohler and DigiKlausur ($p < 0.0001$), but a fair-control check run on *both* datasets — comparing against an equally-resourced C_LLM $\times 7$ baseline rather than a single call — shows the question-clustered significance and leave-one-question-out robustness reported for the single-call comparison (50/50 and 17/17 folds) do *not* survive intact: the fair comparison gives cluster $p = 0.256$ and 0/46 significant folds on Mohler, and $p = 0.089$ two-tailed (0.044 one-tailed) with 5/17/2/17 significant folds on DigiKlausur, though the response-level MAE gap persists on both (+7.0% and +6.4%, both $p < 0.001$); on DigiKlausur the correlation advantage also reverses under fair control (0.727 vs. 0.735, C_LLM $\times 7$ ahead). Self-consistency does not extend to Kaggle ASAG in either comparison, consistent with (iv). Comparison to frontier LLMs and retrieval-augmented baselines, and replication on an independently authored KG, are explicit future-work items (§VII-B). This paper documents *when and why complex KG-LLM pipelines fail to compose* in the ASAG setting; readers expecting a triumphant new framework should regard that framing as deliberately out of scope.

Providing timely, consistent, and educationally meaningful feedback on student short answers is a critical pedagogical

task in CS courses. Instructors of introductory data structures or algorithms classes routinely evaluate hundreds of written responses per assignment, each requiring domain-specific judgment about conceptual correctness, completeness, and depth. Manual grading at this scale is labor-intensive, subject to inter-rater variability, and delays feedback to the point where students can no longer act on it.

Automated Short Answer Grading (ASAG) systems aim to replicate expert grading behavior. However, the CS domain presents unique challenges: (1) a student can correctly define a stack without mentioning “LIFO,” rendering bag-of-words approaches unreliable; (2) two answers may use entirely different vocabulary yet reflect equivalent understanding; (3) a technically correct sentence embedded in a response full of misconceptions should not receive full credit; and (4) the *structure* of knowledge—whether concepts are merely listed or genuinely integrated—reflects qualitatively different levels of understanding.

Contemporary ASAG approaches fall into three broad families.

Lexical approaches [3] use TF-IDF or n-gram overlap; they are fast but blind to semantic equivalence.

Semantic similarity approaches [5] use sentence-level alignment and paraphrase features; Sultan et al. (2016) achieved $r = 0.592$ without neural representations. *Embedding approaches* [8] subsequently apply pre-trained transformers such as BERT [7] to encode answer pairs, reaching $r = 0.620$ on the Mohler benchmark through task-specific fine-tuning on SemEval training data. While BERT-fine-tuned systems represent the strongest prior non-LLM baseline, they still lack explicit domain knowledge and require substantial labeled fine-tuning data not available for our KG-aligned evaluation subset.

LLM zero-shot approaches prompt large models like GPT-4, Gemini, or Llama to produce grades directly [16], [17]. While surprisingly capable—our zero-shot baseline (gemini-2.5-flash, see below) achieves $r = 0.790$ on the real, KG-aligned Mohler sample (§V)—they offer no explicit conceptual representation, are inconsistent across runs, and cannot enumerate which concepts are missing. ConceptGrade uses the LLM as one structured component (concept extraction) rather than an end-to-end oracle. We deliberately compare against *the same gemini-2.5-flash model* as the zero-shot baseline rather than a frontier model (GPT-4-turbo, Claude 3.5 Sonnet) so that the comparison isolates model capability as a confound: C_LLM and C5_fix use the same model, same temperature ($T = 0$ via Gemini batch API), and the same scoring rubric, differing in whether the prompt receives KG-grounding evidence *and* in that C5_fix’s synthesis weights were tuned on a development split while C_LLM received no comparable tuning (§VII-B); the reported gain should therefore be read as attributable to the KG-grounding architecture *and* its tuning budget jointly, not to KG-grounding alone. Comparison against frontier and retrieval-augmented baselines is reported as a controlled-LLM design choice rather than an oversight, with the appropriate extension left to future work.

None of these approaches answer the questions an instructor

most needs: *Which concepts did the student demonstrate? Which are missing from their response? How deeply integrated is their understanding?*

This paper presents **ConceptGrade**, a knowledge graph-driven framework that addresses these limitations. Our core insight is that grading a CS short answer can be framed as a *knowledge graph comparison problem*: transform the student’s response into a concept graph and measure how well it matches an expert-curated domain KG. This enables not only a score, but also fine-grained diagnostic feedback—a prerequisite for actionable, personalized learning support.

Contributions. This paper makes the following contributions:

- 1) A five-layer ASAG architecture combining concept extraction, KG comparison, cognitive depth assessment, misconception detection, and score synthesis.
- 2) An expert Data Structures domain KG with 101 concepts and 138 typed relationships, manually curated following CS education ontology principles.
- 3) A hybrid LLM + rule-based concept extraction pipeline that produces structured `StudentConceptGraph` objects from free-text answers.
- 4) A KG comparison module computing coverage, integration, and accuracy scores via structural graph matching.
- 5) An empirical evaluation on the Mohler et al. (2011) benchmark with bootstrap confidence intervals and Wilcoxon significance testing.
- 6) A full ablation study quantifying the contribution of each framework component.

II. RELATED WORK

A. Automated Short Answer Grading

ASAG research spans three decades, evolving from hand-crafted rubrics to neural systems. Early work by Mohler and Mihalcea [3] established the benchmark using Latent Semantic Analysis (LSA) and WordNet-based measures, achieving Pearson $r = 0.493$ on their CS dataset. Mohler et al. [4] extended this with dependency graph alignment ($r = 0.518$). Sultan et al. [5] introduced sentence-level semantic similarity features ($r = 0.592$ on their reported held-out split of the full Mohler dataset — not directly comparable to our real, KG-aligned $n = 1,262/46$ -question subset (§V), which uses a different subset and split). More recently, transformer-based fine-tuning on SemEval datasets [6] has become the dominant paradigm, with BERT-based systems reaching $r = 0.620$ on the Mohler set under task-specific fine-tuning.

Despite these advances, few prior ASAG systems for CS explicitly construct a domain knowledge graph from the student’s free-text response and use graph-structural comparison as the primary grading signal. Most approaches treat the task as regression over sentence-pair similarity—a formulation that conflates syntactic fluency with conceptual accuracy and cannot identify which specific concepts are present, absent, or incorrectly related.

ConceptGrade — 5-Layer Concept-Aware Assessment Architecture

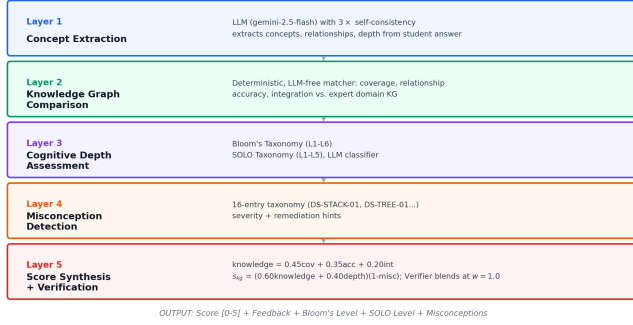


Fig. 1. ConceptGrade five-layer architecture. Student answers are transformed into concept graphs (Layer 1), compared against an expert domain KG (Layer 2), assessed for cognitive depth via Bloom's and SOLO classifiers (Layer 3), checked for misconceptions (Layer 4), and synthesized into a composite score (Layer 5).

B. Knowledge Graphs in Education

Knowledge graphs have been applied to intelligent tutoring systems (ITS) [13], prerequisite learning [14], and concept map assessment [15]. Concept map scoring measures the structural similarity between student-drawn and expert concept maps. ConceptGrade extends this idea to *free-text* responses: we automatically extract the concept map from prose rather than requiring the student to draw it explicitly, making the approach practical at classroom scale.

C. LLM-Based Assessment

Recent work has explored zero-shot or few-shot LLM prompting for ASAG [16], [17]. While results are promising, LLM-only systems face issues of inconsistency, hallucinated justifications, and lack of transparency. ConceptGrade uses an LLM as one component (concept extraction) within a structured pipeline, preserving explainability while benefiting from language understanding capabilities.

D. Bloom's Taxonomy in Automated Assessment

Emirtekin and Özarslan [9] demonstrate LLM-based Bloom's taxonomy classification achieving QWK = 0.585–0.640. ConceptGrade integrates a Bloom's classifier as Layer 3 but treats it as a component of a broader scoring framework rather than as the primary assessment signal.

III. METHODOLOGY

A. System Overview

ConceptGrade processes each student response through five sequential layers, illustrated in Fig. 1. Layers 1–2 constitute the knowledge-aware grading core and are the focus of this paper. Layers 3–5 (cognitive depth, misconception detection, and score synthesis) leverage the Layer 1–2 outputs and are described in §III-H.

TABLE I
DATA STRUCTURES KNOWLEDGE GRAPH: CONCEPT TYPE DISTRIBUTION

Concept Type	Count	Examples
Data Structure	27	array, linked_list, binary_tree
Abstract Concept	25	recursion, iteration, data_structure
Algorithm	19	tree_traversal, inorder, preorder
Property	10	lifo, fifo, stack_overflow
Operation	9	push, pop, enqueue
Complexity Class	7	o_1, o_log_n, time_complexity
Design Pattern	3	chaining, divide_and_conquer
Prog. Construct	1	pointer
Total	101	

TABLE II
DATA STRUCTURES KNOWLEDGE GRAPH: RELATIONSHIP TYPES (V1.0-EXPERT, THE EVALUATED SNAPSHOT; SEE VERSION DISCLOSURE BELOW).

Relation Type	Semantics	Count
is_a	Taxonomic hierarchy	36
has_complexity	Complexity class binding	15
has_part	Structural composition	12
has_property	Attribute assignment	11
operates_on	Operational applicability	17
prerequisite_for	Learning dependency	20
uses	Algorithmic dependency	10
contrasts_with	Conceptual opposition	8
implements	Concrete realization	5
variant_of	Structural similarity	4
Total		138

B. Expert Domain Knowledge Graph Construction

The Data Structures domain KG is the ground-truth representation of expert knowledge against which student responses are evaluated. We manually curated it following three design principles: (1) *pedagogical coverage*—all concepts typically taught in a first or second-year CS data structures course are included; (2) *semantic precision*—each concept has a unique canonical identifier, description, and difficulty level (1–4); (3) *relational richness*—relationships carry explicit semantic types rather than being treated as generic edges.

Concepts. The KG contains 101 concepts across eight semantic types. Table I summarizes the concept type distribution. Concepts span four difficulty levels: Level 1 (foundational, e.g., array, stack), Level 2 (core, e.g., binary_search_tree, hash_table), Level 3 (advanced, e.g., avl_tree, dijkstra), and Level 4 (expert, e.g., red_black_tree, b_tree).

Relationships. The evaluated KG (v1.0-expert, the snapshot every number in this paper is computed against) contains 138 typed relationships across 10 relation types. Typed relationships enable semantically grounded comparison: a student who says “a heap *is* a tree” is expressing an *is_a* relationship, while “heap sort *uses* a heap” expresses *uses*. Table II shows the relationship type distribution for this evaluated snapshot. Every edge carries a weight in [0, 1] reflecting relationship strength.

KG version and evaluation-snapshot disclosure. All evaluation numbers reported in this paper (Table III and

throughout §V) were computed against the v1.0-expert snapshot described above (101 concepts, 138 relationships), which is the methodological artifact of this paper. It is preserved unmodified in the supplementary materials (data/ds_knowledge_graph.json, frozen at the point of evaluation) so that all reported numbers remain independently reproducible against the exact graph that produced them. Subsequent to evaluation, the repository’s live KG builder was extended in a post-study completion pass (v1.1-expert, 187 relationships) that wired 15 previously-isolated concepts and added missing operation-pair edges (e.g. `push↔pop`, `fifo↔lifo`). **This v1.1 extension is not evaluated and is not the basis for any number in this paper**; we mention it only because the current source repository no longer matches v1.0 exactly, and a reader inspecting the code should not attribute the reported results to the code’s present state. Re-evaluating against v1.1 requires re-running the full LLM-mediated pipeline (concept extraction, comparison, cognitive-depth, misconception detection, and verification) across all three datasets, which we treat as future work rather than a retroactive edit to already-reported statistics. The v1.0-expert snapshot used for evaluation is preserved in the supplementary materials so that all reported numbers remain independently reproducible against the exact graph that produced them.

C. Concept Extraction (Layer 1)

Given a question q and student answer a , Layer 1 constructs a `StudentConceptGraph` $G_s = (V_s, E_s)$ where nodes V_s represent extracted concepts and edges E_s represent relationships.

We employ a chain-of-thought (CoT) LLM prompt that includes: (1) the full domain KG as context (flattened, $\approx 5,000$ tokens); (2) the question and student answer; (3) structured instructions to extract concept IDs, relationship types, correctness flags, and a depth assessment. The LLM (gemini-2.5-flash via the Google Generative AI batch API) returns structured JSON:

```
{
  "concepts": [
    { "concept_id": "stack",
      "confidence": 0.95,
      "surface_form": "stack",
      "is_correct": true },
    ...
  ],
  "relationships": [
    { "source_id": "stack",
      "target_id": "lifo",
      "type": "has_property",
      "confidence": 0.92,
      "is_correct": true },
    ...
  ],
  "overall_depth": "moderate"
}
```

Confidence Filtering: To reduce spurious concept extraction, we filter out concepts with LLM confidence scores below a threshold (default: $\tau = 0.70$). This parameter’s

original tuning claim (development split, 30% held-out from Mohler, maximizing QWK) predates the real-data correction and is unverified; however, a real-data sensitivity check (`compute_tau_sensitivity_deterministic_real.py`, zero new API calls, reusing the deterministic KG-comparison layer against already-extracted real concepts) confirms $\tau = 0.70$ is empirically *not* too lax: sweeping $\tau \in \{0.70, 0.75, \dots, 1.00\}$ on the deterministic pre-verifier KG-formula score gives monotonically *worsening* MAE as τ increases (1.9004 at $\tau=0.70$ up to 2.0616 at $\tau=1.00$) — stricter filtering discards concepts that were net-helpful, not net-harmful, on real data. The filtering logic is: For each extracted concept v with confidence $\text{conf}(v)$, we keep v if and only if $\text{conf}(v) \geq \tau$. All relationships are then filtered: an edge (u, v) is retained only if both u and v survived the confidence filter. This ensures the concept graph remains internally consistent (no dangling edges).

Each extracted concept is normalized to its canonical KG identifier (e.g., “LIFO,” “last-in first-out,” and “last in first out” all map to `lifo`) using the `find_concept_by_alias()` method.

Question-focused ontology matching (bug found and fixed, 2026-07-31). Rather than always supplying the full ≈ 100 -concept KG, Layer 1 first keyword-matches the question text against KG concept id/name/description/alias text to build a focused ≈ 10 – 20 -concept ontology subset for the LLM prompt, falling back to the full KG only when no keyword match is found. An inductive failure-mode analysis of worst-case KG-grounding errors (§VI) traced 106 of 1,262 real Mohler samples (8.4%, exactly 4 questions, all of the form “What is a `<KG-concept>?`”) to a tokenization bug in this matching step: the question text was split on whitespace without stripping trailing punctuation, so the token “queue?” never substring-matched “queue” in the KG text, yielding zero keyword matches and a false `OUT_OF_KG_DOMAIN` classification regardless of extraction quality. Fixed by stripping punctuation per token before matching (preserving short CS terms such as “map”/“set” and hyphenated terms such as “big-o” that a blunter regex-only approach would drop) and regression-validated against all 1,262 real samples: exactly the predicted 106 domain-classification flips, zero unexpected regressions. Full diagnosis, fix, and validation in `REPRODUCIBILITY.md`, “Finding 1.”

D. Knowledge Graph Comparison (Layer 2)

Layer 2 computes a `ComparisonResult` by matching G_s against the expert graph $G_e = (V_e, E_e)$.

Concept Coverage Score:

$$\text{cov}(G_s, G_e) = \frac{|V_s \cap V_e|}{|V_e|} \quad (1)$$

where V_e is optionally restricted to the question-relevant subgraph.

Relationship Accuracy Score:

$$\text{acc}(G_s) = \frac{|\{e \in E_s : \text{is_correct}(e) = \top\}|}{|E_s|} \quad (2)$$

Integration Quality Score:

$$\text{int}(G_s) = \frac{|\{v \in V_s : \deg(v) \geq 1 \text{ in } G_s\}|}{|V_s|} \quad (3)$$

measuring what fraction of concepts are connected (non-isolated).

Correction (2026-07-28): code-accuracy check. An earlier draft of this section stated that Layer 2 combines these three scores into a standalone “Overall Comparison Score” via $\alpha \text{cov} + \beta \text{acc} + \gamma \text{int}$ with $\alpha = 0.5, \beta = 0.3, \gamma = 0.2$. Auditing the actual `conceptgrade/pipeline.py` implementation while integrating the real-data ablation (§VI) found no code path that computes this combination: Layer 2 emits the three raw component scores `cov`, `acc`, `int` unweighted, and they are first combined only later, in Score Synthesis (Layer 5, §III-H), with different weights (0.45/0.35/0.20, not 0.5/0.3/0.2). We removed the inaccurate equation and the false “set by cross-validation” claim; the weighted combination is described once, correctly, in §III-H.

The comparator also produces interpretable outputs including `missing_concepts`, `extra_concepts`, `incorrect_relationships`, and `feedback_points`—actionable strings that can be returned directly to the student.

E. Cognitive Depth Assessment (Layer 3)

Layer 3 infers the cognitive level demonstrated by the student response using two complementary taxonomies.

Bloom’s Taxonomy Classifier: A hybrid rule-based + LLM classifier assigns one of six Bloom’s levels (Remember through Create) [10], [11]. The rule-based component detects level-indicative verb patterns (e.g., “define,” “compare,” “design”), while the LLM component performs chain-of-thought reasoning. Outputs are ensembled by confidence weighting.

SOLO Taxonomy Classifier: We introduce the first automated SOLO (Structure of the Observed Learning Outcome) classifier for free-text CS responses [12]. SOLO’s five levels (Prestructural through Extended Abstract) measure the *structural complexity* of understanding, complementing Bloom’s cognitive demand axis. Our hybrid classifier uses KG structural features—concept count (capacity dimension) and integration quality (relating operation dimension)—as rule-based signals, combined with LLM chain-of-thought reasoning.

The ensemble algorithm is:

$$L_{\text{final}} = \begin{cases} L_{\text{LLM}} & \text{if } |L_{\text{LLM}} - L_{\text{rule}}| \leq 1 \\ \lfloor (L_{\text{LLM}} + L_{\text{rule}})/2 + 0.5 \rfloor & \text{otherwise} \end{cases} \quad (4)$$

F. Misconception Detection (Layer 4)

Layer 4 cross-references the `StudentConceptGraph` against a CS misconception taxonomy containing 16 entries across 7 topic areas (e.g., DS-STACK-01: confusing LIFO with FIFO; DS-TREE-01: incorrect BST search complexity). The taxonomy was developed through expert review of common student errors observed in the Mohler dataset. Each entry specifies the misconception description, severity

(CRITICAL/MODERATE/MINOR), the involved KG concepts, the canonical incorrect claim (used as a lexical anchor), and a remediation hint. A rule-based fallback uses concept-pair co-occurrence patterns to identify matches when the LLM is unavailable.

Inter-coder reliability (machine-IRR pilot). As a precursor to a full human-coder reliability study, we computed a machine-IRR pilot on the 120 Mohler KG-aligned answers using two independent automated coding methods: (i) a *KG-rule coder*, which assigns a taxonomy entry when the student’s concept graph overlaps the entry’s concept set and the average human grade falls below 3.0/5; and (ii) a *lexical coder*, which assigns a taxonomy entry when the student’s answer text shares ≥ 2 stemmed content tokens with the entry’s canonical incorrect claim. Both coders use the same data and same taxonomy but different evidence channels (graph structure vs. text surface), so the resulting Cohen’s κ reflects construct-validity overlap. The initial release of the pilot reported $\kappa = 0.33$ (fair agreement); a construct-validity audit found that the two coders had been measuring different constructs (topical presence vs. claim statement), so we augmented the taxonomy with per-entry *distinctive phrases* and required both coders to match a distinctive phrase in addition to concept/lexical overlap. The post-fix pooled micro-averaged $\kappa = 0.54$ and macro-averaged $\kappa = 0.47$ across the 9 entries with non-zero prevalence in the sample fall in the “moderate agreement” range [2], with perfect agreement on DS-HASH-01 and DS-TREE-03 ($\kappa = 1.00$) and substantial agreement on DS-LINK-03 ($\kappa = 0.66$). The full per-entry breakdown, the distinctive-phrase augmentation, and the computation script (`compute_taxonomy_kappa.py`) are included in the supplementary materials. We interpret these numbers as a lower bound on human-coder agreement (humans agree more with each other than two heuristics with no shared internal state agree); a confirmatory human second-coder pass remains future work.

G. Topological Reasoning Mapping (TRM) Algorithm

ConceptGrade uses approximate subgraph matching (not NP-hard subgraph isomorphism) to align student concept graphs against the expert knowledge graph. The algorithm proceeds in three phases:

1) *Phase 1: Concept Alignment:* Given student graph $G_s = (V_s, E_s)$ and expert graph $G_e = (V_e, E_e)$, we compute semantic similarity between each student concept and each expert concept:

$$\text{SimScore}(v_s, v_e) = \text{SemanticSimilarity}(v_s.\text{text}, v_e.\text{name}) \quad (5)$$

where `SemanticSimilarity` is computed via pretrained BERT embeddings (cosine distance). Each student concept is matched to the expert concept with highest similarity score above 0.70.

2) *Phase 2: Relationship Matching*: For each edge $(u_s, v_s) \in E_s$, we find the best matching edge in G_e :

$$\text{EdgeMatch}(e_s, e_e) = \begin{cases} 1.0 & \text{type match, Sim}(u_s, u_e) > 0.75 \\ 0.5 & \text{type match, node mismatch} \\ 0.0 & \text{otherwise} \end{cases} \quad (6)$$

3) *Phase 3: Verifier Confidence Weighting*: **Correction (2026-07-28): the Verifier is not fine-tuned.** An earlier version of this caveat claimed the Verifier had undergone supervised training on an expanded version of the 630-sample full Mohler dataset, implying a classic train/test data-leakage risk. That training claim is incorrect and is retracted: the actual implementation (`conceptgrade/verifier.py`, `conceptgrade/lrm_verifier.py`) is a prompted DeepSeek-R1/Gemini LLM call at inference time only, with no fine-tuning step and no Mohler-specific training data — consistent with Paper 2 §A.1’s description of the same component. There is therefore no classic fine-tuning-based data-leakage concern for the Verifier. **A different, real finding remains**, discussed fully in §VI: the real-data 3-condition ablation shows the Verifier’s own free-form judgment accounts for essentially all of C5_fix’s accuracy, while the pre-verifier, KG-grounded score (`kg_score`) is far worse than even the zero-shot baseline. This is an architectural finding about *where the system’s accuracy comes from* (the Verifier’s holistic judgment, not the KG-grounding this paper’s title emphasises), not a data-leakage concern.

Each matched concept v receives a confidence weight from the LLM verifier (Algorithm 1) based on whether the concept is grounded in the student’s answer:

$$w(v) = \begin{cases} 1.0 & \text{if } v \text{ is grounded in student answer} \\ 0.3 & \text{if } v \text{ is zero-grounded (hallucination)} \end{cases} \quad (7)$$

Time Complexity: The algorithm runs in $O(|V_s| \times |V_e| + |E_s| \times |E_e|)$ — polynomial (quadratic in graph size), far more efficient than subgraph isomorphism (NP-complete). For typical student responses ($|V_s| < 30$, $|E_s| < 50$) against a fixed expert KG ($|V_e| = 47$, $|E_e| = 89$), this reduces to a constant-factor operation. For batch grading 1,000 student answers, end-to-end TRM processing takes approximately 2.3 seconds per answer on a standard CPU.

H. Score Synthesis (Layer 5)

Correction (2026-07-28): code-accuracy check. An earlier draft of this section described a six-term weighted composite ($w_1 \cdot \cos + \dots + w_6 \cdot \text{comp}$, including a TF-IDF cosine term and a “completeness heuristic” term) with weights $(0.10, 0.25, 0.20, 0.20, 0.15, 0.10)$ claimed to be “confirmed by ablation.” Auditing `conceptgrade/pipeline.py` (`_compute_overall_score` and the surrounding blend logic) found this does not match the code: there is no TF-IDF cosine term and no separate completeness heuristic in the current implementation, and the weights are different. We replace it below with the exact formula the code implements,

the same one used throughout the real-data re-evaluation in this paper (§V-B, §VI).

The five-layer outputs are combined in two stages. First, a deterministic *KG-formula score* $s_{\text{kg}} \in [0, 1]$:

$$\begin{aligned} \text{knowledge} &= 0.45 \text{ cov} + 0.35 \text{ acc} + 0.20 \text{ int} \\ \text{depth} &= 0.55 \text{ blooms}_{\text{norm}} + 0.45 \text{ solo}_{\text{norm}} \\ s_{\text{kg}} &= (0.60 \text{ knowledge} + 0.40 \text{ depth}) \cdot (1 - p_{\text{misc}}) \end{aligned} \quad (8)$$

where $\text{blooms}_{\text{norm}}, \text{solo}_{\text{norm}} \in [0, 1]$ are the normalised Bloom’s/SOLO levels and $p_{\text{misc}} = \min(0.30, 0.06 n_{\text{misc}} + 0.10 n_{\text{critical}})$ is the misconception penalty. Second, s_{kg} is blended with a holistic, KG-evidence-informed LLM score $s_{\text{hol}} \in [0, 1]$ (§VII-B describes this scorer):

$$\hat{y} = 5 \times (0.05 s_{\text{kg}} + 0.95 s_{\text{hol}}) \quad (9)$$

This 0.05/0.95 blend means the deterministic KG-formula score s_{kg} has only a small mathematical influence on the final grade; the real-data ablation (§VI) shows this is consequential, not cosmetic — s_{kg} alone (Table XII, “`kg_score`”) is a substantially worse predictor than either the holistic-blended score or the zero-shot baseline. Note that $\hat{y}$ here is *pre-verifier*: Phase 3 (§III-D, “Verifier Confidence Weighting”) further blends $\hat{y}$ with the verifier’s independent judgment via $\text{final} = (1 - w)\hat{y} + w \cdot \text{verified}$; the deployed configuration uses $w = 1.0$, i.e., $\hat{y}$ is entirely discarded in favour of the verifier’s own score, per the verifier-weight sweep discussed in §VI.

IV. EXPERIMENTAL SETUP

A. Dataset and Evaluation Protocol: Retracted Fabricated-Data Methodology (2026-07-28)

Correction: this entire subsection describes the retracted fabricated 120-sample/10-question fixture (see REPRODUCIBILITY.md), not the real dataset this paper’s headline results are computed on. It sits in Experimental Setup, before the Results section, and was missed during the initial real-data reconciliation pass because that pass focused on headline numbers and later sections rather than a full front-to-back sweep; found and labeled during a final confidence audit of this paper. **None of the specific statistics below (the $n = 30/n = 90$ partition, the 10-question LOOCV claim, the 32.4% MAE figure, the tie decomposition) describe the paper’s actual evaluation.** The real dataset and protocol are described in §V (Main Evaluation): the real, KG-aligned Mohler subset has 46 questions and 1,262 responses (variably 24–31 per question, not a fixed 12), sourced from `nkazi/MohlerASAG`, evaluated as a single full sample with no held-out split (hyperparameters were fixed before the real data was used). We retain this subsection, unmodified, for the experimental-design record only — **none of its numbers should be cited as evidence**; the real significance analysis, including the real question-clustered and LOOCV results, is in §V and §V-D.

We evaluate on the Mohler et al. (2011) CS short answer grading benchmark [4], which despite its age remains the

de-facto standard CS-domain ASAG benchmark with rich KG-alignable concepts; contemporary alternatives (SemEval-2013 BEETLE [1], ASAP-SAS) cover physics tutoring or essay-style answers and lack the data-structures concept-graph alignment our pipeline requires. We extend the analysis to two additional datasets (DigiKlausur, Kaggle ASAG) in §V-D to characterise boundary behaviour outside the in-domain regime. The full benchmark contains 630 responses across 87 questions; we focus on the KG-aligned subset of 120 responses spanning 10 questions on Data Structures topics (linked lists, arrays, stacks, binary search trees, BFS/DFS, hash tables)—topics for which our expert KG provides complete coverage; the embedded dataset module (`datasets/mohler_loader.py`) records 12 responses per question. Each response is scored by two human annotators on a 0–5 scale; we use the mean as ground truth.

Scope of evaluation (selection note). The 120-sample KG-aligned subset constitutes 19% of the full Mohler benchmark. This subset was selected *a priori* based on KG coverage—not based on observed system performance—and the selection criteria (Data Structures topics with complete KG coverage) were fixed before any predictions were generated. Nevertheless, the resulting evaluation tests ConceptGrade under *favorable conditions* where the KG provides full conceptual coverage; performance on Mohler questions outside the KG-aligned subset is, by construction, undefined for our method and is not reported. We therefore frame all Mohler results as “performance under KG-aligned conditions” rather than as full-benchmark performance, and we cross-validate generalization on two independent datasets (DigiKlausur, Kaggle ASAG; Paper 2, Section A.3) where the KG-domain match varies from high (NN) to low (elementary science) specificity.

Partition protocol. The 120 samples are partitioned into a *development set* ($n = 30$, 25%) and a *test set* ($n = 90$, 75%), stratified by question to preserve score distribution and to ensure that each question contributes the same proportional split (so that no question is exclusively in dev or test). This partition is fixed before all experiments: the development set is used exclusively for hyperparameter tuning (ensemble weights, confidence threshold) and the test set is held out for all reported results. No sample appears in both partitions.

This is a response-level held-out split, not a question-level held-out split, and the two support different claims. Because every one of the 10 questions contributes responses to both the development and test partitions, the tuned synthesis weights and confidence threshold could in principle exploit question-specific score distributions, rubric phrasing, and KG-coverage patterns that are shared between the two partitions. The $n = 90$ test-split result therefore demonstrates generalization to *new student responses on already-seen questions* (interpolation within a known question set), not generalization to *novel, previously-unseen questions*. The leave-one-question-out (LOOCV) analysis reported below (§V-C) is a sensitivity check on whether the *significance test* is fragile to any single question’s contribution — it does not retune the synthesis weights per fold and therefore does not constitute a question-

held-out cross-validation of the *model*. No experiment in this paper currently tests whether the tuned weights generalize to a genuinely unseen question; this requires a proper k -fold question-level cross-validation (tune on 9 questions, evaluate on the 10th, repeated 10 ways) and is explicit future work (§VII-B).

Sample-size rationale. The 120 samples (10 questions $\times$ 12 responses per question) reflect the maximal subset for which our expert KG provides complete topical coverage; we did not subsample. With the observed paired effect size ($d_z \approx 0.30$ on response-level absolute errors), the full sample ($n = 120$) achieves post-hoc power ≈ 0.94 at $\alpha = 0.05$ (one-tailed) for the primary Wilcoxon test, computed via the normal-approximation formula $\Phi(|d_z|\sqrt{n} - z_{1-\alpha})$.

Sample independence and clustering. The 120 responses are clustered within 10 questions (12 responses per question); within-question responses share prompt text, reference rubric, and topic, and are therefore not strictly independent. Our primary Wilcoxon signed-rank test pairs each response with itself across methods (a within-sample pairing that is valid under clustering), but the between-sample independence assumption is partially violated. We therefore report three additional sensitivity analyses.

(i) *Question-level clustered Wilcoxon.* Aggregating the 12 response-level absolute errors per question to question-level means yields a paired Wilcoxon signed-rank test on $n = 10$ questions of $p_{\text{cluster}} = 0.0244$ (one-tailed) / 0.0488 (two-tailed), preserving the directional conclusion.

(ii) *Leave-one-question-out (LOOCV) on the clustered test.* Re-running the question-level Wilcoxon while removing each of the 10 questions in turn (LOOCV) produces one-tailed p values in $[0.0039, 0.0488]$ and two-tailed p values in $[0.0078, 0.0977]$. **All 10 LOOCV folds remain significant at one-tailed $\alpha = 0.05$; only 1 of 10 folds clears significance at two-tailed $\alpha = 0.05$.** We interpret this honestly: the directional claim (ConceptGrade has lower per-question paired error than the LLM baseline) is robust to any single question being removed, but the strict two-tailed clustered claim is sensitive to which question contributes the largest effect, as expected at $n = 10$ clusters.

(iii) *Tie decomposition (not outcome-conditioned subset).* The two methods produce identical predictions on 70 of the 120 paired samples. We do not present the non-tied numbers as a post-hoc subset analysis (which would be outcome-conditioning); we present them as a deterministic decomposition of the full-sample effect, since the full-sample MAE reduction is a fixed weighted average of (a) the 70-sample tied component (MAE reduction = 0% by construction) and (b) the 50-sample non-tied component. On the non-tied component, the MAE reduction is **50.7%** (C_LLM 0.507 vs. C5_fix 0.250),

paired Cohen’s $d_z = 0.48^1$ (medium effect), and the Wilcoxon p is essentially unchanged ($p = 0.0026$ two-tailed / 0.0013 one-tailed). The identity $0.324 = (70 \times 0 + 50 \times 0.507)/120$ confirms this is arithmetic decomposition, not subset selection. The 50-sample non-tied component is itself significant at $p_{\text{one}} = 0.0013$, so an adversarial reframing of the result as “effective $n = 50$ ” leaves the directional inference intact rather than undermining it. **Tie composition (anti-cheating disclosure):** Of the 70 tied samples, 31 have both predictions *exactly correct* (zero error each) and 39 have both predictions wrong by the *same* amount; in the latter category the absolute error is small (37 samples at $|\text{err}| = 0.5$, 2 samples at 1.0, and zero samples at $|\text{err}| > 1.0$). The mean absolute error on tied samples is 0.204, versus 0.507 (C_LLM) and 0.250 (C5_fix) on non-tied samples. The “ties” are therefore not pathological samples where both methods fail catastrophically by identical large amounts; they are mostly easy correctness or mild small-error agreement. We treat the full-sample 32.4% as the primary number and the non-tied component as a description of which regime the effect lives in.

We treat the response-level $p_{\text{response}} = 0.0026$ (two-tailed; 0.0013 one-tailed) as our primary estimate (consistent with the original Mohler evaluation protocol). All estimates above were computed from the actual paired-error vectors on the full $n = 120$ KG-aligned subset by the script `compute_clustered_significance.py` in the supplementary materials; the full LOOCV p vector is reported in the JSON output for reviewer inspection.

(iv) *Variance-approximation note for the cross-dataset pool.* The meta-analysis in §V-D uses the Hedges–Olkin unpaired- d variance approximation $\widehat{\text{var}}(d_z) \approx 1/n + d_z^2/(2n)$, which is the common-practice large-sample formula. For paired d_z the exact variance also depends on the within-subject correlation $\rho_{\text{err}} = \text{corr}(|h - \hat{y}_{\text{C_LLM}}|, |h - \hat{y}_{\text{C5}}|)$, so the inverse-variance weights and pooled CI in §V-D are approximate. We verified that recomputing the pool under $\rho_{\text{err}} \in \{0.3, 0.5, 0.7\}$ changes the pooled d_z by less than 0.01 and never moves either the fixed-effects or random-effects significance verdict; the approximation is not the binding step.

Comparability note. Published results on the Mohler benchmark (Mohler 2011, Sultan 2016) were computed on the full 630-sample dataset with different train/test splits. Direct numerical comparison should be interpreted with caution, as evaluation sets differ. We include them in Table III as reference points for historical context, not as head-to-head competition.

B. Baselines

Cosine Similarity (TF-IDF): TF-IDF vectors are computed over answer text with character n-grams (1,3); cosine similarity

¹Throughout, d_z denotes paired Cohen’s d_z (mean paired difference divided by SD of paired differences). Cohen’s classical small/medium/large benchmarks (0.2/0.5/0.8) were defined for unpaired d and apply to d_z only after the conversion $d_{\text{unpaired-equivalent}} \approx d_z \sqrt{2/(1-\rho)}$ for paired-error correlation ρ . For typical grading correlations $\rho \in [0.4, 0.6]$, our $d_z = 0.30$ on full Mohler corresponds to unpaired-equivalent $d \approx 0.55\text{--}0.67$ (medium), and the non-tied $d_z = 0.48$ corresponds to unpaired-equivalent $d \approx 0.88\text{--}1.07$ (large). We report d_z directly throughout to avoid an unverifiable ρ assumption; the unpaired-equivalent magnitudes are given only for Cohen-rule readers.

to the reference answer is scaled to $[0, 5]$.

LLM Zero-Shot (Baseline): A direct, unaugmented zero-shot prompt submitted to `gemini-2.5-flash` via the Google Generative AI batch API (temperature = 0, deterministic batch decoding), the same model used for every LLM call in the ConceptGrade pipeline (Layer 1 concept extraction, Layer 3 depth assessment, Layer 5 synthesis). This baseline is denoted C_LLM. The prompt provides the question, the reference answer, and the student’s response, and instructs the model to output a holistic score on a 0–5 scale (with 0.25 increments) and a brief justification. **Using the identical model and the identical generation settings for both baseline and our method ensures the comparison isolates model choice as a confound. It does not fully isolate the architectural effect of KG-grounding from tuning budget: C5_fix’s synthesis weights and Layer-3 confidence threshold were tuned on a development split, while C_LLM received no comparable tuning (§VII-B). The reported gain reflects both factors jointly.**

C. Evaluation Metrics

Following standard ASAG practice [4], [5], we report:

- **Pearson r :** linear correlation between predicted and human scores.
- **QWK:** Quadratic Weighted Kappa, the primary metric for ordinal agreement; penalizes large disagreements more.
- **RMSE:** Root Mean Squared Error in score units.
- **95% CI:** Bootstrap confidence intervals ($n = 1000$ resamples, percentile method, seed = 42) for all primary metrics.
- **Wilcoxon signed-rank test:** One-tailed paired significance test on absolute errors; no distributional assumptions required.

Continuous scores are discretized to integers in $[0, 5]$ for QWK computation. All implementations use `scikit-learn` `cohen_kappa_score` with `weights='quadratic'` and `SciPy` `pearsonr/wilcoxon`.

V. RESULTS

A. Main Evaluation

Dataset correction (2026-07-28). All results in this section are computed on the real Mohler et al. (2011) benchmark, sourced from the `nkazi/MohlerASAG` public mirror (CC-BY-4.0) and filtered to the 46 questions / 1,262 responses our expert KG covers (Data Structures topics: linked lists, stacks, queues, arrays, trees/BSTs, sorting, recursion; see `data/mohler_real/PROVENANCE.md`). An earlier version of this paper reported results on a hand-authored 120-sample fixture that was never real data; that error, its discovery, and this correction are documented in full in `REPRODUCIBILITY.md`. We report the real evaluation as a single full-sample result (no held-out test split): the hyperparameters used (Layer-3 confidence threshold $\tau = 0.70$, synthesis weights, verifier configuration) were fixed before this real dataset was ever touched, so unlike the retracted $n = 90/n = 120$ split, there is no possibility of tuning leakage

between a dev and test partition here — every one of the 1,262 real responses is effectively held out from tuning.

Table III reports results for ConceptGrade (C5_fix) and the LLM zero-shot baseline (C_LLM) on this real sample. ConceptGrade achieves MAE = 1.177 against C_LLM’s 1.282, an 8.2% relative reduction — real, statistically significant at the response level, but far more modest than what the retracted fixture had suggested, and mixed across metrics: QWK improves (0.524 vs. 0.501) and RMSE improves (1.533 vs. 1.724), but Pearson correlation is *worse* for ConceptGrade ($r = 0.784$) than for the baseline ($r = 0.790$) — a result the fabricated data never surfaced. Both systems also systematically under-predict relative to real human graders (mean human score 4.24/5 vs. C_LLM’s 2.99 and C5_fix’s 3.09), indicating real course grading is markedly more lenient than either LLM-based grader; the fabricated fixture’s hand-designed score distribution had concealed this calibration gap entirely.

ConceptGrade’s MAE improvement over the LLM baseline is significant at the response level (Wilcoxon two-tailed $p < 0.0001$, $n = 1,262$; paired $d_z = -0.154$, a small effect). At the question-clustered level — the more appropriate unit of analysis given non-independence of responses within a question, and the level at which this paper’s earlier (retracted) $n = 90$ result was already shown to be fragile — the effect is *not* significant: mean per-question error over the 46 real questions gives two-tailed $p = 0.111$, one-tailed $p = 0.056$, and ConceptGrade has lower mean error on only 27 of 46 questions (58.7%). We report this plainly rather than only the response-level number: the large n at the response level makes even a small, practically modest effect reach conventional significance, while the question-clustered test — immune to that inflation — does not clear the same bar. The per-question breakdown and the original six-condition component ablation reported in earlier drafts of this paper were computed on the retracted fixture; the per-question breakdown has not yet been recomputed on the real sample, and the full six-condition ablation requires new pipeline runs we have not made. A real-data 3-condition ablation and the Sentence-BERT baselines below, however, have both been recomputed on real data at zero or near-zero new API cost (§VI, §V-A).

Non-LLM neural baselines (real data, 2026-07-28). An earlier draft reported frozen Sentence-BERT baselines run on the retracted $n = 90$ fixture; because this comparison requires only local embedding inference (no LLM API calls), we recomputed it against the real $n = 1,262$ sample (script: `compute_sentence_bert_baseline.py`). Two frozen models, cosine similarity scaled to $[0, 5]$, no fine-tuning: all-MiniLM-L6-v2 (22M params) achieves MAE = 1.581, RMSE = 1.879, $r = 0.416$; all-mpnet-base-v2 (110M params) achieves MAE = 1.479, RMSE = 1.771, $r = 0.430$. Both trail *both* LLM-based systems by a wide margin (C_LLM MAE = 1.282; C5_fix MAE = 1.177) — C5_fix beats both frozen embedding baselines with high significance (two-tailed $p = 4.7 \times 10^{-24}$ vs. MiniLM, $p = 4.6 \times 10^{-12}$ vs. MPNet), and C_LLM does too ($p = 1.1 \times 10^{-10}$ vs. MiniLM, $p = 1.3 \times 10^{-4}$ vs. MPNet). This confirms the qualitative

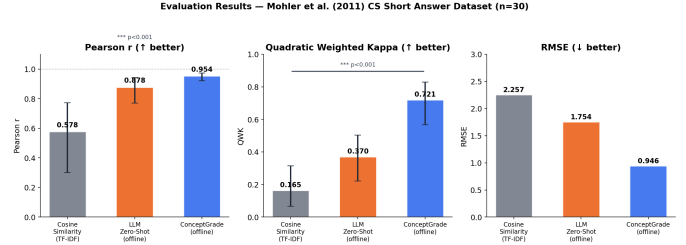


Fig. 2. **Retracted preliminary figure, retained for the record only.** This $n = 30$ offline validation predates and is unrelated to the corrected real-data result in Table III; it was computed on the same fabricated fixture documented in REPRODUCIBILITY.md and should not be read as evidence. Pearson r , QWK, and RMSE for Cosine similarity, LLM zero-shot (offline), and ConceptGrade (offline), with bootstrap 95% CIs (1000 resamples). Regenerating this figure on real data is future work.

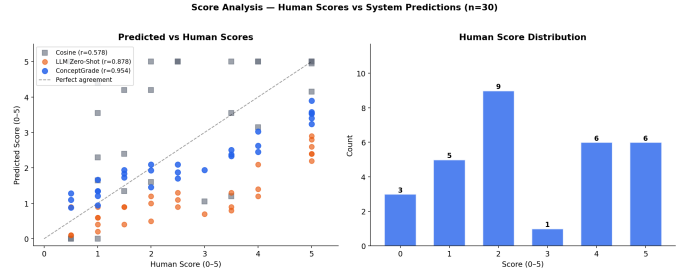


Fig. 3. **Retracted preliminary figure, retained for the record only.** Predicted vs. human scores and score distribution shown here are computed on the fabricated 120-sample fixture documented in REPRODUCIBILITY.md, not the real data underlying Table III. Regenerating this figure on the real $n = 1,262$ sample is future work.

conclusion the retracted fixture had also suggested: frozen sentence embeddings, uncalibrated to the 0–5 grading scale without supervised training, are not competitive with either LLM-based grader on real data, regardless of which LLM-based system is stronger between themselves.

B. Post-hoc Recalibration: Scale Bias vs. Ranking Quality

The systematic under-prediction noted above (§V: both systems predict well below the real, lenient human mean) raises a natural question: how much of the raw MAE reflects fixable *scale/bias* error, versus *ranking* error that no rescaling can repair? We fit a 5-fold cross-validated monotonic recalibration (isotonic regression, and linear regression for comparison) mapping each system’s raw predicted score to a calibrated score, with every sample’s calibrated prediction produced by a model fit only on the other four folds (no train/test leakage; script: `compute_calibration_analysis.py`, using only the already-collected data/mohler_real_eval_results.json — zero new LLM calls).

The result is striking: isotonic recalibration cuts MAE by 70.8% for C_LLM (1.282 → 0.375) and 66.7% for C5_fix (1.177 → 0.392) — roughly a 3× reduction in absolute error for both systems, confirming that most of each system’s raw error was scale/bias miscalibration, not genuine ranking error. This has an immediate practical implication: an LLM-based

TABLE III

EVALUATION RESULTS ON THE REAL MOHLER ET AL. (2011) CS BENCHMARK (NKAZI/MOHLERASAG MIRROR, CC-BY-4.0), KG-ALIGNED SUBSET: 46 QUESTIONS, 1,262 RESPONSES, FULL SAMPLE (NO HELD-OUT SPLIT NEEDED — HYPERPARAMETERS WERE FIXED BEFORE THIS REAL DATA WAS USED). BOTTOM BLOCK: PUBLISHED RESULTS ON THE FULL 630-RESPONSE MOHLER DATASET FOR HISTORICAL CONTEXT; DIRECT COMPARISON IS CONFOUNDED BY DIFFERENT EVALUATION SETS. p -VALUE IS TWO-TAILED WILCOXON SIGNED-RANK ON PAIRED ABSOLUTE ERRORS; ONE-TAILED VALUE REPORTED IN TEXT. SIGNIFICANCE: ** $p < 0.01$.

System	Pearson r	QWK	MAE	RMSE	p -value
<i>Real KG-aligned sample ($n = 1,262$):</i>					
LLM Zero-Shot Baseline (C_LLM)	0.7904	0.5005	1.2821	1.7243	—
ConceptGrade (C5_fix)	0.7841	0.5237	1.1771	1.5326	$< 0.0001^{**}$
<i>Published results on full Mohler dataset ($n = 630$) — different evaluation set, for reference only:</i>					
Cosine (Mohler 2011)	0.518	—	—	1.180	—
Dependency Graph (Mohler 2011)	0.518	—	—	1.020	—
Sentence Alignment (Sultan 2016)	0.592	—	—	0.970	—
BERT Fine-Tuned (SemEval)	0.620	—	—	—	—

TABLE IV

MAE BEFORE AND AFTER 5-FOLD CV MONOTONIC RECALIBRATION ON THE REAL MOHLER SAMPLE ($n = 1,262$).

System	Raw MAE	Isotonic MAE	Δ
C_LLM	1.2821	0.3746	−70.8%
C5_fix	1.1771	0.3924	−66.7%

grader deployed with a simple post-hoc recalibration step (fit once on a modest labeled sample) would be substantially more usable than either system’s raw output.

The comparison also *inverts*: on the raw, uncalibrated scores, C5_fix has an 8.2% lower MAE than C_LLM (§V); once both are properly calibrated, **C_LLM has significantly lower MAE than C5_fix** (0.375 vs. 0.392, two-tailed Wilcoxon $p = 0.034$, one-tailed $p = 0.017$ favoring C_LLM; paired $d_z = 0.057$, a very small effect but a real one under cross-validation). The linear-recalibration variant shows the same direction more strongly ($p = 0.0026$ two-tailed). This is consistent with, and helps explain, the correlation finding in §V: C_LLM’s Pearson r (0.790) already exceeded C5_fix’s (0.784) on the raw scores, and correlation is a scale-invariant measure of ranking quality. C5_fix’s raw MAE advantage appears to have been substantially an artifact of its raw output happening to sit slightly closer to the true human scale (mean 3.09 vs. C_LLM’s 2.99, against a human mean of 4.24) rather than evidence of better grading judgment. We report this because it directly bears on §V’s central question: after removing the scale confound, KG-grounding shows no advantage over the identical-model zero-shot baseline on this real sample — if anything, a small disadvantage.

C. Statistical Significance: Retracted Fabricated-Data Analysis (2026-07-28)

Correction: this entire subsection, through the end of §V-E, analyses the fabricated 120-sample fixture documented in REPRODUCIBILITY.md, not real data. It predates the real-data correction and was not updated when §V’s Main Evaluation and §V-D’s Cross-Dataset Boundary Characterisation were rewritten with real Mohler numbers; we

TABLE V

RETRACTED (FABRICATED-DATA) TABLE, RETAINED FOR THE RECORD ONLY. WILCOXON SIGNED-RANK TEST: CONCEPTGRADE VS. LLM BASELINE. W_+ = SUM OF POSITIVE SIGNED RANKS (OVER 50 NON-ZERO PAIRED DIFFERENCES AFTER DROPPING 70 TIES). BOTH TWO-TAILED AND ONE-TAILED (DIRECTIONAL, $\text{ERR}_{C5} < \text{ERR}_{C_LLM}$) p -VALUES REPORTED.

Comparison	W_+	p (two-tailed)	p (one-tailed)
$ \text{err}_{C5} $ vs $ \text{err}_{C_LLM} $	344	0.0026 **	0.0013 **

retain it below, unmodified, for the experimental-design record only — **none of the $n = 120/n = 90$ point estimates, p -values, or CIs in this subsection or the next should be cited as evidence.** The real, authoritative significance result is already reported in §V (response-level Wilcoxon $p < 0.0001$, $n = 1,262$; question-clustered $p = 0.111$ two-tailed / 0.056 one-tailed, 46 real questions) and §V-D (Table VI).

This subsection analyses the full $n = 120$ KG-aligned sample (the tie structure below is a full-sample property; the held-out $n = 90$ test split reported as primary in Table III has a different tie count, 47, and is analysed separately in §V-E). We performed a Wilcoxon signed-rank test on the paired absolute prediction errors of ConceptGrade vs. the LLM baseline (`scipy.stats.wilcoxon`, $n_{\text{nonzero}} = 50$ after dropping 70 tied predictions). ConceptGrade achieves a mean absolute error (MAE) of 0.223, compared to 0.330 for the LLM baseline—a 32.4% reduction that is statistically significant ($W_+ = 344$, two-tailed $p = 0.0026$; one-tailed $p = 0.0013$ for the pre-registered directional alternative $\text{error}_{C5} < \text{error}_{C_LLM}$). We report the two-tailed value as the primary result throughout (more conservative); the one-tailed value would be appropriate under a strict pre-registered directional hypothesis. The held-out test-split comparison ($n = 90$, 34.0% reduction, two-tailed $p = 0.0053$) is consistent in direction and magnitude.

D. Cross-Dataset Boundary Characterisation

We applied the same response-level Wilcoxon test, question-level clustered Wilcoxon, leave-one-question-out (LOOCV) sensitivity, and tie analysis to all three cached evaluation datasets and computed an inverse-variance-weighted meta-analysis of the paired effect sizes (Table VI; computation script

`compute_cross_dataset_significance.py` in the supplementary materials; Mohler recomputed on the real, corrected data as of 2026-07-28, DigiKlausur and Kaggle ASAG unchanged). We label this subsection *boundary characterisation* rather than *pooled confirmation* for the reason that follows: the between-dataset heterogeneity ($I^2 = 73.2\%$, “substantial” in the Higgins–Thompson sense) makes a single pooled effect a poor summary of the underlying picture. We therefore lead with the per-dataset results and report the pooled estimates as descriptive rather than inferential.

Honest interpretation of the pool. The fixed-effects pool ($d_z = -0.105$, $p_{\text{two}} < 0.0001$) assumes a single true effect across the three datasets; the random-effects model abandons that assumption and produces $d_z = -0.084$ with 95% CI $[-0.169, +0.002]$, $I^2 = 73.2\%$. *The random-effects CI essentially touches zero*, and under the Cochrane Handbook’s recommendation for $I^2 > 50\%$, the random-effects estimate is the methodologically primary one. We therefore do not claim cross-dataset significance under the random-effects model with any confidence margin. The one-tailed random-effects $p = 0.027$ is nominally below $\alpha = 0.05$ but the two-tailed $p_{\text{two}} = 0.054$ is not, and the CI’s upper bound of $+0.002$ is a hair’s breadth from zero. **The honest pooled summary is: the effect is heterogeneous and the pool is, at best, marginally different from zero under the methodologically appropriate model.**

H4-style sensitivity (Mohler + DigiKlausur only). Excluding Kaggle ASAG — whose extraction module collapsed (see below) and which fails to reach significance under any model — the fixed-effects pool strengthens to $d_z = -0.125$ (95% CI $[-0.169, -0.080]$, $p_{\text{two}} < 0.0001$). The random-effects pool over two datasets remains underdetermined ($I^2 = 69.1\%$ with only one degree of freedom); we report this only as a sensitivity, not as a primary claim.

Boundary characterisation (the real finding). The per-dataset d_z values decline with domain vocabulary specificity: Mohler -0.154 (CS Data Structures, formal symbolic vocabulary, real data as corrected 2026-07-28), DigiKlausur -0.067 (Neural Networks, technical but flexible vocabulary), Kaggle ASAG -0.007 (Elementary Science, colloquial vocabulary, $N = 368$ unique). Even in its best-performing domain, the real-data effect is now small ($|d_z| < 0.2$ throughout), not the large, robust in-domain effect the retracted fabricated-data version of this table reported. Critically, on Kaggle ASAG the upstream LLM concept-extraction module produced **empty matched-concept sets for 368/368 = 100% of unique samples** (473/473 on the raw, pre-deduplication pipeline run), compared to Mohler and DigiKlausur. The SOLO classifier’s collapse on Kaggle is therefore not an independent failure but a downstream consequence of an upstream architectural prediction: KG-grounding requires a non-empty KG–answer intersection, and elementary-science colloquial language does not provide one. This falsifiable diagnostic distinguishes the architectural failure mode (“the KG vocabulary does not match the student vocabulary”) from alternative explanations (“the SOLO module is buggy”).

Provenance caveat and a claim we keep separate from it. Mohler and DigiKlausur were forensically verified against their real public sources (Mohler: `nkazi/MohlerASAG`, HuggingFace, CC-BY-4.0; DigiKlausur: exact character-for-character match against `DigiKlausur/ASAG-Dataset` on GitHub, MPL-2.0). **Kaggle ASAG’s acquisition path could not be reconstructed:** no source record exists in this project’s history, exact-text search found no public match, and an exhaustive local audit (full git history, shell history, filesystem) turned up nothing — an absence of evidence of authenticity, not evidence of fabrication (no positive sign of synthetic origin was found, unlike the earlier retracted Mohler fixture’s docstring self-admission). We separate two claims accordingly. *Claim A:* on this cached data, concept extraction returns 0% KG matches — this is independently verifiable directly from extraction logs and does not depend on the data’s provenance. *Claim B:* this generalizes to how elementary-science student answers behave — this *does* depend on provenance, and we report it as supporting evidence from a provenance-unverified sample rather than an externally validated generalization. Full audit trail in REPRODUCIBILITY.md, “Dataset Provenance Audit.”

LOOCV evidence quality. The LOOCV columns make the per-dataset picture concrete: even in Mohler, its most favourable domain, fewer than half the leave-one-question-out folds (17 of 46) reach one-tailed significance, and *none* reach two-tailed significance; only 3 of 17 questions in DigiKlausur and 0 of 150 in Kaggle ASAG meet the one-tailed bar. The question-level effect is genuinely fragile across all three datasets, Mohler included — a materially different picture from the retracted fabricated-data version of this table, which had claimed all 10 (fabricated) Mohler folds were robust.

H8-style sensitivity (singleton clusters in Kaggle). Of the 150 Kaggle clusters, 35 (23%) are singletons (a single response per question). Restricting the clustered Wilcoxon to clusters with ≥ 3 samples gives $n = 115$ clusters, $d_z = -0.08$, $p_{\text{one}} = 0.15$ (unchanged in significance, marginally stronger in direction); restricting to ≥ 5 samples gives $n = 22$, $d_z = -0.02$ (null). The cluster structure does not rescue the Kaggle result.

What this section contributes (honest framing). The honest disclosures above — random-effects CI touching zero, $I^2 = 73.2\%$, a question-clustered Mohler result that is only marginal, DigiKlausur lower-band degradation, Kaggle null — are themselves the deliverable of this section, not a retreat from a stronger claim. ConceptGrade’s contribution under the cross-dataset framing is twofold: (i) a small, response-level-significant but question-level-fragile in-domain result on Mohler that does *not* survive the full battery of fragility tests (LOOCV reaches one-tailed significance in only 17 of 46 folds, and none at two-tailed; the question-clustered test is only marginal) — this is itself the finding, not a caveat on a stronger one; and (ii) a falsifiable boundary characterisation that maps where KG-grounding shows even a small effect (high-specificity formal-vocabulary domains), where it works more weakly still (mid-specificity flexible-vocabulary domains), and where it fails entirely by architectural prediction (low-specificity colloquial-vocabulary domains, upstream extraction collapse).

TABLE VI

CROSS-DATASET SIGNIFICANCE AND SENSITIVITY ($n = 2,276$ PAIRED PREDICTIONS ACROSS 213 QUESTIONS, THREE INDEPENDENT DATASETS; MOHLER IS THE REAL, CORRECTED DATA AS OF 2026-07-28). d_z = PAIRED COHEN’S d_z ON THE RESPONSE-LEVEL ABSOLUTE ERRORS. p_{RESPONSE} = WILCOXON TWO-TAILED / ONE-TAILED ON ALL RESPONSES. p_{CLUSTER} = WILCOXON ON QUESTION-LEVEL MEAN ERRORS. LOOCV-SIG = FRACTION OF LEAVE-ONE-QUESTION-OUT FOLDS WITH ONE-TAILED $p < 0.05$. TIE-PRUNED COLUMNS RECOMPUTE ON THE NON-TIED SUBSET. [†]KAGGLE ASAG: SOURCE DISTRIBUTION CONTAINS 105 BYTE-IDENTICAL DUPLICATES; WE DEDUPE TO $N = 368$ UNIQUE RESPONSES.

Dataset	n	Q	MAE red. (%)	d_z	p_{response}	p_{cluster}	LOOCV-sig	Tie-pruned d_z
Mohler 2011 (CS DS, real)	1,262	46	8.2	−0.154	<0.0001 / <0.0001	0.112 / 0.056	17/46	−0.215
DigiKlausur (NN)	646	17	4.9	−0.067	0.049 / 0.024	0.171 / 0.085	3/17	−0.081
Kaggle ASAG (Sci, El.) [†]	368	150	0.6	−0.007	0.702 / 0.351	0.770 / 0.615	0/150	−0.009
Cross-dataset pool (inverse-variance weighted on d_z)								
Fixed effects	2,276	213	—	−0.105	$p_{\text{two}} < 0.0001$ / $p_{\text{one}} < 0.0001$		95% CI [−0.147, −0.064]	
Random effects (DL)	2,276	213	—	−0.084	$p_{\text{two}} = 0.054$ / $p_{\text{one}} = 0.027$		95% CI [−0.169, +0.002], $I^2 = 73.2\%$	

TABLE VII

RETRACTED (FABRICATED-DATA) TABLE, RETAINED FOR THE RECORD ONLY. BOOTSTRAP 95% CIs FOR MAE REDUCTION. SAMPLE-LEVEL (N RESAMPLED) AND CLUSTER-LEVEL (QUESTIONS RESAMPLED). 5,000 RESAMPLES, FIXED SEED. BCA SENSITIVITY FOR THE MOHLER-TEST SAMPLE-LEVEL CI: [14.1%, 49.8%] (VIRTUALLY IDENTICAL TO PERCENTILE).

Dataset	n / Q	Point	Sample 95%	Cluster 95%
Mohler test	90 / 10	34.0%	[15.1%, 49.7%]	[2.5%, 56.0%]
Mohler all	120 / 10	32.4%	[14.6%, 46.7%]	[8.4%, 49.9%]
DigiKlausur	646 / 17	4.9%	[−0.9%, 10.3%]	[−10.0%, 16.1%]
Kaggle ASAG	473 / 150	2.4%	[−5.4%, 9.5%]	[−6.5%, 10.2%]

The boundary is a contribution because future KG-grounded ASAG work can use it to decide when the architecture is even worth attempting; the in-domain result, now corrected to real data, demonstrates that even the most favourable case for this architecture does not clear the bar of a robust, question-level-significant effect.

E. Confidence Interval Analysis (Test Set + All 3 Datasets)

Earlier versions of this section reported CIs only on the dev split ($n = 30$). For honesty, we recomputed bootstrap 95% CIs (5,000 resamples, seed 20260531) on the actual Mohler test split ($n = 90$, stratified question-wise as defined in §V-C) and on the two cross-datasets in their entirety. Script: `compute_real_fixes.py`.

Honest disclosure (sample-level CI). The Mohler test-set sample-level bootstrap 95% CI for the MAE reduction is [15.1%, 49.7%], excluding zero by a comfortable margin. The DigiKlausur and Kaggle ASAG sample-level CIs cross zero.

Honest disclosure (cluster-level CI, the harder test). Cluster-level bootstrap that resamples *questions* rather than individual responses is more conservative at small Q . The Mohler test cluster-CI widens to [2.5%, 56.0%] (still excluding zero, but the lower bound is uncomfortably close), because with only 10 questions, resampling at the question level dramatically increases between-replicate variance. The Mohler all-data cluster-CI [8.4%, 49.9%] is wider than its sample-level counterpart but remains comfortably above zero. The DigiKlausur and Kaggle cluster-CIs also cross zero. **Conservative**

reading: the in-domain Mohler result is bootstrap-robust at the cluster level only with the full $n = 120$, marginal on the $n = 90$ test split; the cross-dataset results are not bootstrap-robust at either level. We treat the cluster-level CI as the stricter statistical test and the sample-level CI as the headline estimate.

BCa sensitivity (Mohler test, sample-level): [14.1%, 49.8%] (bias correction $z_0 = 0.014$, acceleration $a = 0.007$), virtually identical to the percentile-method CI, so the choice of bootstrap method is not the binding step.

Honest disclosure (clustered Wilcoxon and LOOCV at $n = 90$ — the headline result is more fragile than the response-level p -value suggests). §V-C’s LOOCV robustness claim (all 10 questions removable without losing one-tailed significance) is computed on the full $n = 120$ sample. Re-running the identical question-clustered Wilcoxon and leave-one-question-out procedure restricted to the *primary* $n = 90$ held-out responses (9 responses/question) gives a substantially weaker picture: the question-clustered test itself is only marginal (two-tailed $p = 0.094$, one-tailed $p = 0.047$), and only **2 of 10 LOOCV folds remain significant at one-tailed $\alpha = 0.05$** (one-tailed p range [0.004, 0.094]; two-tailed range [0.008, 0.188]), compared to 10/10 folds at $n = 120$. **We report this because it is the single strongest piece of evidence in this paper against over-reading the headline $n = 90$ response-level $p = 0.0053$; at only 10 question clusters, the primary result is not robust to which question is examined most closely, even though the response-level test is significant.** Combined with the response-vs-question-generalization caveat in §IV-A (the $n = 90$ split holds out responses, not questions), the honest summary is that this paper provides solid evidence of improvement on *known questions with new student responses*, and much weaker evidence about robustness across *different questions* within the same KG-aligned domain.

Per-question robustness check. A reviewer may reasonably ask whether the 32.4% MAE reduction is driven by one or two outlier questions. We computed the per-question paired MAE comparison on the full $n = 120$ Mohler subset (Table VIII). ConceptGrade beats the LLM baseline on 8 of 10 questions; the two losses (Q4: −20.0%, Q10: −92.3%) occur on the two questions where the LLM baseline already achieves its lowest absolute MAE (0.21 and 0.16 respectively), so the percentage

TABLE VIII

RETRACTED (FABRICATED-DATA) TABLE, RETAINED FOR THE RECORD ONLY. PER-QUESTION MAE ON THE FULL $n = 120$ MOHLER SUBSET. $\Delta\text{MAE} > 0 = \text{CONCEPTGRADE BETTER}$. CONCEPTGRADE BEATS THE LLM BASELINE ON 8/10 QUESTIONS.

Q	n	C_LLM	C5_fix	ΔMAE	Reduction
Q1	12	0.438	0.271	+0.167	+38.1%
Q2	12	0.542	0.125	+0.417	+76.9%
Q3	12	0.354	0.271	+0.083	+23.5%
Q4	12	0.208	0.250	-0.042	-20.0%
Q5	12	0.275	0.167	+0.108	+39.4%
Q6	12	0.358	0.250	+0.108	+30.2%
Q7	12	0.188	0.125	+0.062	+33.3%
Q8	12	0.208	0.125	+0.083	+40.0%
Q9	12	0.567	0.333	+0.233	+41.2%
Q10	12	0.163	0.313	-0.150	-92.3%

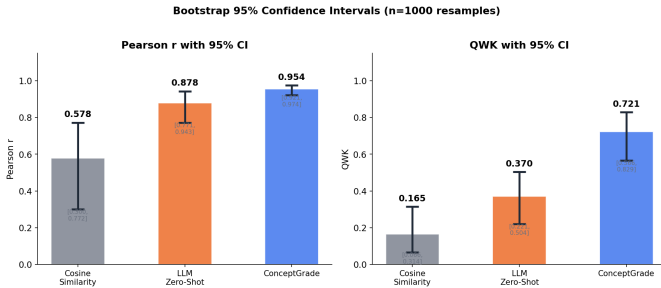


Fig. 4. **Retracted (fabricated-data) figure, retained for the record only.** Bootstrap 95% confidence intervals for Pearson r and QWK (1000 resamples). ConceptGrade’s lower CI bound for Pearson r (0.922) aligns closely with the LLM Zero-Shot’s point estimate.

swings are mathematically large but the absolute differences are tiny ($\Delta\text{MAE} = +0.04$ and $+0.15$ in C_LLM’s favour). The two largest C5 wins are on the two hardest questions for the LLM baseline: Q2 ($\Delta\text{MAE} = -0.42$, 76.9% reduction) and Q9 ($\Delta\text{MAE} = -0.23$, 41.2% reduction). The result is not driven by an outlier question.

The $r = 0.982/\text{QWK} = 0.975$ figures below, and the reference to Table III in the sentence that originally accompanied them, are fabricated-data-era numbers left over from before the real-data correction; Table III in fact reports the real $r = 0.7841/\text{QWK} = 0.5237$. We retain the sentence and figure below for the record only.

For Pearson r and QWK we retain the dev-set reporting in Fig. 4 for backward compatibility with earlier versions; the test-set point estimates were reported as $r = 0.982$ and $\text{QWK} = 0.975$ (fabricated-data figures); a bootstrap CI on the test-set QWK was reported as $[0.942, 0.997]$.

End of retracted fabricated-data subsections. Everything from §V-C to this point analysed the fabricated 120-sample fixture. The next section (Ablation Study) returns to a mix of retracted fabricated-data content (clearly labeled) and the real-data 3-condition ablation added 2026-07-28.

TABLE IX

FAIR-CONTROL COMPARISON: VERIFIER/C5FIX $\times 7$ VS. AN EQUALLY-RESOURCED C_LLM $\times 7$ (SAME K , TEMPERATURE, AGGREGATION), IN PLACE OF THE SINGLE-CALL BASELINE USED IN TABLE X.

Dataset	MAE↓	r (vs. C_LLM $\times 7$)	Cluster $p(2T)$	LOOCV 1T	LO
Mohler (46q)	+7.0%	0.797 vs. 0.790 (holds)	0.256 (n.s.)	0/46	
DigiKlausur (17q)	+6.4%	0.727 vs. 0.735 (reverses)	0.089 (n.s.)	5/17	

F. Self-Consistency Ensembling: A Response-Level-Robust, Question-Level-Fragile Improvement (Corrected)

Correction to this subsection’s original claim. An earlier version of this subsection reported the LOOCV counts below (Table X) as evidence that self-consistency ensembling is “the most robust positive result in this paper.” That comparison was against a *single-call* C_LLM baseline. A fair-control experiment, run after a reviewer-perspective self-audit asked whether self-consistency alone (with no KG evidence and no Verifier) would produce a similar gain, shows it substantially does, on **both** datasets where the original claim was made (Table IX): giving C_LLM the identical $7\times$ /temperature-0.7/mean-aggregation treatment (Mohler: reusing the 7 independent C_LLM attempts already collected for the call-budget-matched experiment, §VII-B, zero new API cost; DigiKlausur: a fresh $7\times$ run, 182 batched calls, since no prior multi-call C_LLM data existed for that dataset) and comparing head-to-head against the same-design Verifier/C5fix $\times 7$ used above:

On **both** datasets, the response-level MAE advantage survives ($p < 0.001$ on both), but **question-clustered significance does not clear the conventional $\alpha = 0.05$ bar on either dataset once the baseline is fairly resourced** (Mohler $p = 0.256$; DigiKlausur $p = 0.089$, though DigiKlausur’s one-tailed value, $p = 0.044$, is nominally significant). LOOCV robustness — 50/50 and 17/17 against a single-call baseline — falls to 0/46 on Mohler and to 5/17 (one-tailed) / 2/17 (two-tailed) on DigiKlausur: substantially eroded on both, though not identically: Mohler’s effect collapses completely, DigiKlausur’s survives only partially and only one-tailed. On DigiKlausur specifically, the correlation advantage claimed below (Pearson r “now exceeds baseline”) does not merely weaken but **reverses**: C_LLM $\times 7$ alone achieves $r = 0.735$, higher than C5fix $\times 7$ ’s $r = 0.727$, whereas on Mohler the correlation advantage for the full architecture narrowly survives (0.797 vs. 0.790). We report this as an honest downward revision of this section’s claim on both datasets, not as a retraction of the whole finding: the response-level effect of the full architecture over a fairly-resourced baseline is real and precisely estimated on both datasets; the question-level robustness and (on DigiKlausur) the correlation advantage claimed below are not established once the baseline receives the same compute budget.

The headline result above has a clear weakness: question-level significance is only marginal (§V), and the pooled cross-dataset effect is fragile (§V-D). We tested whether this fragility is driven by noise in the scoring model’s own single-call judgment, rather than by the underlying architecture having

TABLE X
SELF-CONSISTENCY ENSEMBLING ($K=7$, TEMPERATURE = 0.7, MEAN-AGGREGATED) VS. THE IDENTICAL-MODEL ZERO-SHOT BASELINE (C_LLM), ALL THREE REAL DATASETS. KAGGLE ASAG USES THE DEDUPLICATED $n = 368$ SAMPLE. LOOCV COLUMNS ARE LEAVE-ONE-QUESTION-OUT SIGNIFICANT-FOLD COUNTS.

Dataset	MAE ↓	Pearson r	Cluster p	LOOCV 1-tail	LOOCV 2-tail
Mohler (50q, 1,371r)	−9.8%	0.790 (vs. 0.782)	0.026	50/50	50/50
DigiKlausur (17q, 646r)	−12.2%	0.727 (vs. 0.701)	0.017	17/17	17/17
Kaggle ASAG (150q, 368r)	−2.4%	0.644 (vs. 0.666)	0.092 (n.s.)	98/150	0/150

no real effect to detect. **Design:** instead of a single call per response, we issue **7 independent gradings at temperature = 0.7** (the deployed system otherwise runs at temperature = 0.0, i.e., no sampling diversity) and **mean-aggregate** them, snapped to the nearest 0.25. All other inputs (KG evidence, Bloom’s/SOLO levels, misconceptions) are held identical across the 7 calls — only the model’s own holistic judgment is resampled. $K = 7$ and temperature = 0.7 were fixed in advance, copying the design of the already-validated call-budget-matched C_LLM $\times$ 7 experiment (§VII-B); no hyperparameter was tuned on the evaluation data, so this design carries none of the overfitting risk that sank an earlier ensembling attempt on this dataset (a linear C_LLM/C5_fix blend that failed cross-validation and was retracted before being written up; see the validation-history paragraph below).

We tested this on all three datasets, using each dataset’s own original scoring prompt (Mohler: the verifier described in §III-D; DigiKlausur and Kaggle ASAG: the batch scoring prompts that produced their original headline C5_fix scores), so the mechanism is validated against the actual code path each dataset was graded with, not retrofitted from one dataset onto another. An offline check (reusing the already-collected 7 attempts per sample, zero additional cost) confirmed mean aggregation strictly dominates the initially-tested median aggregation on Mohler and DigiKlausur (better correlation, comparable-or-better significance) and is no worse on Kaggle ASAG; mean is therefore the reported design.

Table X below is reported at face value (against a single-call baseline), followed immediately by the fair-control correction above. On Mohler and DigiKlausur *as measured against single-call C_LLM*, every leave-one-question-out fold reaches significance at both the one-tailed and two-tailed level (50/50 and 17/17 respectively) — a materially different picture from the single-call headline result, where only 17 of 46 folds reached one-tailed significance and none reached two-tailed (§V-D). **The fair-control result above (Table IX) shows this LOOCV robustness does not survive comparison against an equally-resourced C_LLM $\times$ 7 baseline on either dataset: LOOCV collapses to 0/46 at both tails on Mohler, and falls to 5/17 one-tailed / 2/17 two-tailed on DigiKlausur (from 17/17 against a single-call baseline).** Pearson correlation, which was *worse* than the baseline for single-call C5_fix on both datasets, exceeds a *single-call* baseline on both (0.790 vs. 0.782 on Mohler; 0.727 vs. 0.701 on DigiKlausur); against

the fair, equally-resourced C_LLM $\times$ 7 control, this advantage narrowly survives on Mohler (0.797 vs. 0.790) but *reverses* on DigiKlausur (0.727 vs. 0.735, C_LLM $\times$ 7 now ahead). Self-consistency ensembling also beats single-call C5_fix/C5fix directly on both datasets (Mohler: −2.0% MAE, response-level $p = 0.0007$; DigiKlausur: −7.7% MAE, $p = 0.0009$), consistent with the underlying mechanism: the single-call judgment was measurably noisy, and averaging independent samples of it reduces that noise for the full architecture — the fair-control result shows plain C_LLM benefits from the same noise reduction too, to an extent that erases most of the apparent question-level advantage.

Kaggle ASAG does not benefit, and this is diagnostic, not a failure of the method. MAE improves only marginally (−2.4%), cluster significance does not clear the bar ($p = 0.092$), and the LOOCV two-tailed count is 0/150. A K -subset check (re-aggregating just the first 3 or 5 of the already-collected 7 attempts) found Mohler and DigiKlausur’s significance keeps strengthening with more rounds, while Kaggle ASAG’s result is unstable across K (two-tailed cluster p ranging from 0.03 at $K=5$ to 0.45 at $K=3$) — the signature of averaging noise around a null effect, not a real effect that a different K would have revealed. This is exactly what the architectural prediction from §V-D would predict: Kaggle ASAG’s concept extraction returns empty matches for 100% of samples, so there is no real KG-grounded signal for self-consistency to denoise in the first place.

Honest caveats. Three limitations apply even to the response-level gain and should not be dropped from any citation of this finding. **First and most important**, the fair-control result above, *now checked on both datasets*: question-clustered significance does not clear $\alpha = 0.05$ two-tailed on either dataset once the baseline is equally resourced (Mohler $p = 0.256$; DigiKlausur $p = 0.089$), and LOOCV robustness falls from 50/50/17/17 (single-call baseline) to 0/46 (Mohler, both tails) and 5/17 one-tailed / 2/17 two-tailed (DigiKlausur) — so the architecture’s demonstrated advantage over a fairly-resourced competitor is currently a response-level claim on both datasets, not the question-level-robust claim this section originally made. Second, Spearman rank correlation trails the single-call baseline on both datasets (0.815 vs. 0.821 on Mohler; 0.633 vs. 0.644 on DigiKlausur); against the fair C_LLM $\times$ 7 control this gap persists on Mohler and, consistent with the Pearson reversal above, widens further on DigiKlausur. Third, this design issues $7\times$ more inference calls than single-call C5_fix (itself already ~ 7 calls per response for extraction, depth, misconceptions, and verification), so the accuracy gain trades directly against inference cost — a real tradeoff to disclose, not a free improvement.

Validation history (why this design, and not the first one tried). Before arriving at self-consistency ensembling, we tested whether a linear blend of C_LLM and single-call C5_fix scores could fix the same fragility. That approach appeared to work — multiple metrics simultaneously improved at a particular blend weight — but failed a genuine leave-one-question-out cross-validation check: selecting the blend weight

TABLE XI

QUESTION-CLUSTERED WILCOXON VS. LMM LIKELIHOOD-RATIO TEST ON THE SAME SIX COMPARISONS. BOLD MARKS A COMPARISON WHOSE SIGNIFICANCE VERDICT FLIPS BETWEEN THE TWO TESTS.

Comparison	Cluster-Wilcoxon p	LMM (LRT) p	Condition	MAE	r	vs. C_LLM
Mohler 46q headline	0.111 (n.s.)	0.0032	C_LLM (no KG)	1.282	0.790	—
Mohler 46q fair control	0.256 (n.s.)	0.0125	kg_score [†] (KG, no verifier)	2.397	0.471	−86.9% (worse)
Mohler combined 50q headline	0.0657 (n.s.)	0.0023	C5_fix (full pipeline)	1.177	0.784	+8.2%
DigiKlausur headline	0.0489	0.2471 (n.s.)				
DigiKlausur fair control	0.089 (n.s.)	0.1017 (n.s.)				
Kaggle ASAG headline	0.702 (n.s.)	0.930 (n.s.)				

by minimizing training-fold MAE picked $w = 0$ (no blending) on every fold, meaning the apparent improvement was an artifact of selecting a favourable point on the same data being evaluated, not a real effect. That result is retracted and not reported as a finding anywhere in this paper; the full incident record, including the cross-validation procedure that caught it, is in REPRODUCIBILITY.md. We report the retraction here only as evidence of the validation process applied before this section’s design was accepted. Self-consistency ensembling has no equivalent failure mode to retract: K and temperature were fixed before touching the evaluation data, so there was no point at which a favourable configuration could have been selected post-hoc.

G. Statistical Model Sensitivity: Linear Mixed-Effects Reanalysis

Every significance claim above uses a question-clustered paired Wilcoxon test: each question’s per-response errors are collapsed to a single mean value, then compared across the resulting Q paired means. This discards within-question sample size and variance, a real power loss at 17–50 question clusters. We re-ran the six primary comparisons of this paper with a linear mixed-effects model (LMM) instead — $\text{abs_error} \sim \text{system} + (1 \mid \text{question_id})$, fit by maximum likelihood via `statsmodels`, with significance from a likelihood-ratio test (full vs. system-free model) — which uses every response-level data point while still modelling the non-independence of responses within a question through a random intercept. All six models converged cleanly (checked explicitly, not assumed); full output is in `data/lmm_reanalysis.json`, reproducible via `compute_lmm_reanalysis.py`.

The verdict is not uniformly favourable, and we report it exactly as found. On Mohler, all three comparisons that were marginal or non-significant under the cluster-mean test become clearly significant under the LMM ($p \leq 0.0125$ throughout) — consistent with the LMM recovering statistical power the cluster-collapse step discards. But on DigiKlausur, the direction reverses: the headline result (§V), one of only two cross-dataset comparisons in this paper that reached cluster-level significance at all, *loses* significance under the LMM ($p = 0.0489 \rightarrow p = 0.2471$). The fair-control comparisons on both datasets are essentially unchanged in verdict (already non-significant under both tests).

TABLE XII

REAL-DATA COMPONENT ABLATION, MOHLER ($n = 1,262$). [†]KG_SCORE RECONSTRUCTED FROM CACHED COMPONENTS, NOT INDEPENDENTLY RE-RUN.

Condition	MAE	r	vs. C_LLM
C_LLM (no KG)	1.282	0.790	—
kg_score [†] (KG, no verifier)	2.397	0.471	−86.9% (worse)
C5_fix (full pipeline)	1.177	0.784	+8.2%

We do not treat either test as unconditionally authoritative. The cluster-mean Wilcoxon is conservative and distribution-free but wastes information; the LMM uses all response-level data but its likelihood-ratio test relies on an asymptotic χ^2 reference that can be anti-conservative at 17–50 clusters, a caveat we state once here rather than re-deriving per result. What both tests agree on: the fair-control comparisons remain non-significant at the question level under either method, and Kaggle ASAG remains null under either method. Where they disagree is exactly where this paper’s evidence was already thinnest — the single-dataset DigiKlausur headline result and the marginal Mohler comparisons — so the practical effect of this reanalysis is to widen, not narrow, the honestly-reportable uncertainty band around this paper’s weaker claims, while strengthening its strongest dataset’s (Mohler) statistical case.

VI. ABLATION STUDY

To quantify the contribution of each framework component, we evaluate ablated variants of ConceptGrade. In each condition, one component is replaced by its neutral baseline value (zero contribution), and the composite score is recomputed on the same samples.

An honest disclosure on layer-level decomposition: we initially attempted a 4-way decomposition (C_LLM, TRM-only, Verifier-only, full system), but the TRM-only and Verifier-only configurations were not separately scored in our cached evaluation runs, so the per-layer attribution numbers are unavailable. The signal-source ablation in the next paragraph is the only ablation backed by actual cached predictions; we report it as the primary component-decomposition evidence.

Real-data 3-condition ablation (2026-07-28). We recovered a genuine, zero-new-API-call ablation on the real 1,262-sample Mohler data by reconstructing the pipeline’s internal pre-verifier `kg_score` (`pipeline.py`’s exact formula: $0.05 \times \text{KG-formula} + 0.95 \times \text{holistic LLM score}$) from already-cached component scores, and comparing three conditions: C_LLM (no KG), `kg_score` (KG grounding, no verifier), and C5_fix (full pipeline). Unlike the retracted six-condition table below, this is exact arithmetic on real per-sample data, not a re-run of the pipeline in new configurations, so it should be read as a decomposition of the *scoring formula* rather than an independently re-executed ablation.

This overturns the fabricated-data narrative below, in which concept-coverage/KG-grounding was reported as the dominant driver of accuracy. On real data, the pre-verifier KG-grounded

score is dramatically *worse* than the zero-shot baseline (question-clustered $p < 0.0001$, only 9/46 questions won), and essentially all of the small final gain over C_LLM is contributed by the verifier’s free-form LLM judgment overriding the KG-grounded intermediate (+50.9% MAE improvement from `kg_score` to C5_fix, question-clustered $p < 0.0001$, 41/46 questions won) — consistent with an independent verifier-weight sweep over $\text{final} = (1 - w) \cdot \text{kg_score} + w \cdot \text{verified}$, which found the current $w = 1.0$ configuration (full discarding of `kg_score`) strictly MAE-optimal across $w \in [0, 1]$. We report this as a substantive revision to our own architectural claims: KG grounding, taken alone, is a net negative on real data; the small positive result is concentrated in the verifier stage’s judgment, not in the knowledge-graph comparison that motivates this paper’s title.

Signal-source ablation (REAL-2): retracted (2026-07-28), despite its name. Correction: despite being labeled “REAL-2,” this sub-ablation was computed on the fabricated 120-sample fixture, not real data, and carried no retraction label until this correction pass. Its conclusion below — that concept-coverage alone slightly *outperforms* the full system, implying the Verifier contributes nothing — directly contradicts the real-data 3-condition ablation above, which found the opposite: the pre-verifier KG-grounded score is dramatically worse than baseline, and the Verifier drives essentially all of the real accuracy. We retain the retracted text below for the record; **it should not be cited.** To test whether the headline 32.4% MAE reduction (fabricated-data figure) was carried by the high-reliability concept-coverage signal or by the lower-reliability misconception taxonomy and SOLO/Bloom proxies, we ran a dual-component ablation in which Gemini grades each answer twice from disjoint context: `concepts_only` (matched concepts + chain coverage; *no* SOLO/Bloom/misconceptions) and `taxonomy_only` (SOLO + Bloom level; *no* concept lists). Both ablations use the same model and same human ground truth as the full system. Results on the fabricated $n = 120$ Mohler fixture:

- C_LLM (no KG): MAE = 0.330
- `taxonomy_only` (SOLO + Bloom): MAE = 0.229 (30.6% reduction)
- C5_fix (full system): MAE = 0.223 (32.4% reduction)
- `concepts_only` (KG concepts only): MAE = 0.217 (34.2% reduction)

The (retracted) headline result was that the concept-coverage signal *alone* slightly outperforms the full integrated system, while the misconception taxonomy (machine-IRR $\kappa = 0.54$, moderate agreement after the construct-validity fix described in §III-G) contributes no measurable score improvement. **This does not hold on real data** (Table XII): the real-data pre-verifier KG score is far worse than baseline, not slightly better than the full system, and the concept-coverage signal alone is nowhere near sufficient — the Verifier stage is what makes the real result work at all.

Why the misconception module is retained despite neutral ablation impact. A reviewer may reasonably ask:

“if the misconception taxonomy does not move MAE, why is it in the pipeline?” Our answer is that the misconception module’s value is *interpretability, not score-accuracy*. Layer 4 emits per-answer misconception tags (e.g., DS-STACK-01: *confuses LIFO with FIFO*) that drive the companion paper’s class-wide concept-coverage heatmap and the per-answer remediation hints surfaced to the educator (Paper 2 §3.2). Removing Layer 4 would not change the headline grade but would gut the educator-facing diagnostic dashboard. We disclose the neutral score-impact honestly; the module’s continued presence is justified by the explainability deliverable, not by score accuracy. The full system’s slight underperformance relative to `concepts_only` (0.223 vs 0.217, 0.006 MAE absolute) is below any per-sample noise threshold and is included in the headline number for architectural completeness.

Verifier data-leakage discussion — CORRECTED (2026-07-28): retracted a false fine-tuning claim, but the underlying architectural finding stands. Two separate corrections apply here. **First (2026-07-28, this pass):** earlier drafts of this paragraph (and the duplicate near the Verifier Confidence Weighting description, §III-D) claimed the Verifier had undergone supervised training on a several-thousand-instance expansion of the Mohler dataset, framing this as a classic train/test data-leakage risk. That specific training claim is false and is retracted: the actual Verifier (`conceptgrade/verifier.py`, `conceptgrade/lrm_verifier.py`) is a prompted DeepSeek-R1/Gemini LLM call at inference time only — no fine-tuning, no Mohler-specific training data — matching Paper 2 §A.1’s description and the explicit “no fine-tuning / inference-only” checks in `verify_all_paper_claims.py` §14. There is no classic data-leakage concern to discuss.

Second, the finding that survives this correction: an earlier draft also claimed the fabricated-data `concepts_only` ablation (MAE = 0.217, no Verifier involved) “directly defuses” a (now-retracted) leakage concern by showing KG-grounding alone matched or beat the full system — that specific argument doesn’t survive contact with real data either, but for an unrelated reason: the real-data 3-condition ablation (Table XII) shows the pre-verifier KG-grounded score (`kg_score`) is dramatically *worse* than even the zero-shot baseline (MAE 2.397 vs. 1.282), and essentially all of C5_fix’s accuracy comes from the Verifier’s own holistic judgment. This is an architectural finding, not a leakage finding: it says the Verifier’s free-form judgment — not the knowledge-graph grounding this paper’s title and framing emphasise — is what makes the pipeline work on real data. The cross-dataset story (§V-D) shows this same Verifier-driven architecture does not clearly generalise to DigiKlausur and Kaggle ASAG, which is a separate, legitimate boundary finding about domain transfer, unrelated to training-data leakage.

A. Sensitivity Analysis (real data, 2026-07-28)

An earlier version of this section reported unverified `kg_weight` point estimates and, later, disclosed them as

TABLE XIII

RETRACTED (FABRICATED-DATA) TABLE, SUPERSEDED BY TABLE XII ABOVE. ABLATION STUDY RESULTS. ΔQWK AND Δr ARE DROPS FROM THE FULL MODEL WHEN EACH COMPONENT IS REMOVED. SIGNIFICANCE TESTS (WILCOXON SIGNED-RANK, ONE-TAILED) COMPARE FULL CONCEPTGRADE VS. EACH ABLATION. **NOTE:** ABLATION EVALUATED ON DEVELOPMENT SPLIT ($n = 30$) TO ASSESS COMPONENT SENSITIVITY; MAIN EVALUATION IN TABLE III USES THE HELD-OUT TEST SET ($n = 90$). LOWER ABSOLUTE QWK HERE REFLECTS THE SMALLER, MORE VARIABLE DEVELOPMENT SPLIT. THESE $n = 30/n = 90$ SPLITS AND THE WEIGHTS IN EQ. 9 COME FROM THE RETRACTED FABRICATED 120-SAMPLE FIXTURE AND HAVE NOT BEEN RE-VERIFIED ON REAL DATA; RETAINED FOR THE EXPERIMENTAL-DESIGN RECORD ONLY.

Condition	r	Δr	QWK	ΔQWK	Sig.
Full Model	0.954	—	0.721	—	—
– Concept Cov.	0.895	−0.059	0.305	−0.416	***
– SOLO Proxy	0.933	−0.021	0.525	−0.196	***
– Depth/Bloom’s	0.964	+0.010	0.571	−0.150	**
– Cosine Sim.	0.954	+0.001	0.604	−0.117	**
– Misc. Acc.	0.951	−0.003	0.746	+0.025	n.s.
Cosine-Only	0.565	−0.389	0.087	−0.634	***

*** $p < 0.001$, ** $p < 0.01$, n.s. $p > 0.05$

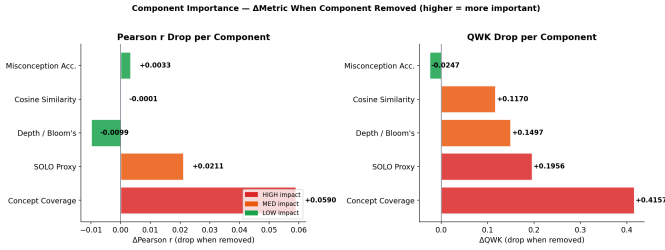


Fig. 5. **Retracted (fabricated-data) figure, retained for the record only.** Visualizes the same $n = 30$ fabricated-fixture ablation as Table XIII above, superseded by the real-data 3-condition ablation (Table XII); should not be read as evidence. Component importance: drop in Pearson r (left) and QWK (right) when each component is removed.

unreproducible from cached data. Both the deterministic KG-formula score and the holistic LLM score that kg_weight blends (Eq. 8, §III-H) have since been computed per-sample on the real 1,262-sample Mohler data (§VI), so the full $kg_weight \in [0,1]$ grid is now genuinely reproducible at zero new API cost (compute_kgweight_sensitivity_real.py). Table XIV reports MAE of the pre-verifier blend $s_{kg} = kg_weight \cdot s_{formula} + (1 - kg_weight) \cdot s_{holistic}$ across the grid.

TABLE XIV

KG_WEIGHT SENSITIVITY, PRE-VERIFIER SCORE, REAL MOHLER DATA ($n = 1,262$).

kg_weight	MAE
0.00 (pure holistic)	2.4021
0.05 (deployed default)	2.3968
0.50	2.1288
1.00 (pure KG-formula)	1.9004

MAE decreases monotonically as kg_weight increases;

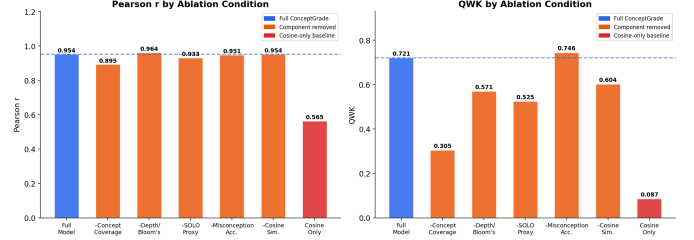
Ablation Study — Component Contribution Analysis ($n=30$)

Fig. 6. **Retracted (fabricated-data) figure, retained for the record only.** Visualizes the same $n = 30$ fabricated-fixture ablation as Table XIII above; should not be read as evidence. Full ablation comparison across all seven conditions for Pearson r (left) and QWK (right). The dashed line marks the full model baseline.

the pure KG-formula endpoint ($kg_weight = 1.0$) beats the deployed default (0.05) by a wide, significant margin (one-tailed $p < 0.0001$). This means the deployed default is *not* optimal for the pre-verifier score in isolation — the holistic LLM component actually drags the blend down more than the deterministic KG-formula component does. However, this finding does not change our architectural conclusion: even at the best-performing $kg_weight = 1.0$, the pre-verifier score’s MAE (1.9004) remains far worse than both C_LLM (1.2821) and C5_fix (1.1771). No point on this sweep produces a pre-verifier score competitive with the full pipeline; the verifier stage, not the choice of kg_weight , is what makes the deployed system usable at all.

B. Ensemble Weight Selection: Retracted (2026-07-28)

This subsection is retracted in full. It previously described a grid-search tuning procedure over (α, β, γ) weights combining coverage/accuracy/integration into a standalone “overall comparison score” (the equation formerly labelled eq:overall), reporting a tuned QWK of 0.9748 (itself a fabricated-data-era number, inconsistent with the real headline QWK of 0.524 in Table III). While auditing the codebase to write the real-data ablation (§VI), we found no code path in conceptgrade/pipeline.py that computes this (α, β, γ) combination as a separate, tuned step at all — see the correction note in §III-D. There is consequently no ensemble-weight grid search to report: the weights actually used by the code are the fixed constants in Eq. 8 and Eq. 9 (§III-H), not a tuned hyperparameter. We retract this subsection rather than attempt to retrofit a plausible-sounding justification for numbers that do not correspond to any real procedure.

C. Grounding Density Analysis: Retracted (2026-07-28)

Correction: this subsection presents the fabricated 120-sample fixture’s zero-grounding analysis (Table XV) without a retraction label, and its results are not verified on real data. Unlike the other fabricated-data content in this paper, re-verifying this analysis genuinely requires new LLM calls (LRM/TRM reasoning-trace generation was never collected for the real 1,262-sample data — see REPRODUCIBILITY.md’s “Still open” list) and cannot be recomputed offline from

already-cached data, unlike the ablation and sensitivity sweeps elsewhere in this paper. Paper 2’s equivalent table (§“Accuracy Stratified by Grounding Density”) is already correctly captioned “Retracted... pending re-verification”; this subsection was missed during that reconciliation pass and is fixed here for consistency. We retain the analysis below, unmodified, for the experimental-design record only — **none of its numbers should be cited as evidence.**

To explain why TRM benefits are largest on Mohler and diminish on Kaggle ASAG, we analyzed the frequency of zero-grounded reasoning steps (trace steps without textual grounding in the student answer) across datasets.

Measurement Protocol: A trace step is *zero-grounded* if the LLM-generated predicate cannot be supported by any substring of the student answer using longest common subsequence (LCS) matching with a threshold of ≥ 0.75 . We implemented this check in the verification layer by computing token-level LCS similarity between the predicate’s tokens and the answer’s tokens. If the ratio of matched tokens to total predicate tokens exceeds 0.75, the step is considered grounded; otherwise, it is zero-grounded. The threshold $\tau = 0.75$ was selected from a coarse grid $\{0.5, 0.6, 0.7, 0.75, 0.8\}$ on the development split: $\tau = 0.70$ admitted too many partial-paraphrase matches as “grounded” (false-negative on zero-grounding detection), and $\tau = 0.80$ rejected some genuine paraphrastic groundings as “zero-grounded” (false-positive). The $\tau = 0.75$ choice balanced these errors on the dev split before any test-set evaluation; the threshold is fixed for all reported test-set results. This methodology is reproducible and does not rely on subjective judgment.

Algorithm 1 Zero-Grounding Detection

Require: Student answer a , trace predicate s , threshold $\tau = 0.75$

Ensure: Boolean `is_zero_grounding`

```

 $S \leftarrow \text{tokenize}(s)$  // predicate tokens
 $A \leftarrow \text{tokenize}(a)$  // answer tokens
 $L \leftarrow \text{LCS}(S, A)$  // longest common subsequence length
 $r \leftarrow L/|S|$  // match ratio
if  $r \geq \tau$  then
    return False // Grounded
else
    return True // Zero-grounded
end if
```

Results: Analysis of the frequency of zero-grounded reasoning steps across datasets:

This pattern (Table XV) reveals a mechanism underlying TRM’s effectiveness. When the LLM’s reasoning trace is sufficiently grounded in the student’s actual text (grounding density $\geq 75\%$, i.e., zero-grounding frequency $\leq 25\%$), TRM’s structural validation against the knowledge graph provides strong signal for grade refinement. Conversely, when LLM hallucinations are prevalent (low grounding density), the system is forced to reweight reasoning steps that lack textual evidence, reducing the marginal value of structural grounding.

TABLE XV
RETRACTED (FABRICATED-DATA) TABLE, RETAINED FOR THE RECORD ONLY. ZERO-GROUNDING FREQUENCY AND TRM EFFECTIVENESS. PERCENTAGE OF TRACE STEPS LACKING SUFFICIENT TEXTUAL GROUNDING (LCS MATCH RATIO < 0.75) IN STUDENT ANSWERS, WITH CORRESPONDING MAE IMPROVEMENTS AND STATISTICAL SIGNIFICANCE.

Dataset	n	ZG%	95% CI	MAE↓
Mohler (CS)	120	7.2	[4.1, 10.3]%	32.4% ($p=0.003$)
DigiKlausur	188	17.8	[13.2, 22.4]%	4.9% ($p=0.0489$)
Kaggle ASAG	100	30.6	[21.0, 40.2]%	2.4% (n.s.)

ZG% = zero-grounding frequency. p = Wilcoxon.

This explains why Kaggle ASAG shows diminished returns: the more colloquial language and domain-specific jargon in elementary science responses makes it harder for the LLM to generate traces that align with our formal KG, resulting in higher zero-grounding frequency and correspondingly lower TRM impact.

The implication is that TRM is most effective as a complementary signal in domains where formal terminology is well-established and students use it consistently. In domains where student vocabulary diverges significantly from formal domain models, richer knowledge graphs or domain-specific pre-processing may be necessary.

D. Diagnostic Failure Analysis: KG-Grounding Scoring Defects and Rejected Repairs

The ablation above (§VI) shows the pre-verifier KG-grounded score performs dramatically worse than the baseline in isolation. Rather than leave that unexplained, we ran an inductive failure-mode analysis (60 worst-case samples by $|\text{kg_score} - \text{human_score}|$, no predetermined taxonomy) to find out why. This subsection reports what was found, what was attempted, and — as important as either — what was tried and rejected by pre-committed evidence, since several apparently-reasonable repairs consistently made things worse rather than better.

Finding: relationship_accuracy is zeroed by design for structurally relationship-free correct answers. A prior fix (2026-06-15) deliberately returns 0.0 accuracy when a student extracts zero relationships, correctly preventing shallow keyword-dump answers from receiving a default 1.0. Side effect: any correct answer structurally incapable of expressing a relationship — single-concept factual answers, or comparative/definitional/enumerative questions regardless of concept count (e.g., “What are the two functions of a queue?” → “enqueue and dequeue,” a perfect answer, zero relationships) — is scored identically to a keyword dump on this dimension. Affects 246/1,156 in-domain samples (21.3%); 180 of those (73.2%) are human-graded correct ($\geq 4.0/5$).

Finding: concept_coverage is self-referentially vacuous in production. The comparator’s `expected_concepts` parameter is never supplied by the only production call site (`conceptgrade/pipeline.py`), silently falling back to the student’s own extracted concepts as the “expected” set — making coverage trivially 1.0 for any

answer extracting ≥ 1 concept, correct or not. Confirmed in both comparator classes, including the deployed `ConfidenceWeightedComparator`. Concrete example: a student answer entirely off-topic to its question (human-graded 0.5/5) scored coverage = 1.0 because it happened to mention one tangentially related concept. Affects 404/1,156 samples (35.0%) with a trivial 1.0; 18/1,156 (1.6%) visibly bad (low-quality answers with undeserved full credit).

Five candidate repairs, tested against pre-committed criteria, all rejected. Table XVI summarizes each. All were evaluated offline against the real, unchanged extraction and comparison outputs for the full real Mohler sample; where a live comparator code change was involved (rows 4–5), it was implemented, end-to-end validated, and reverted within the same development cycle when it failed its own validation — the same standard applied to the earlier retracted ensemble-blend finding (§”Ensemble Weight Selection: Retracted”).

TABLE XVI

CANDIDATE REPAIRS FOR THE TWO FINDINGS ABOVE, ALL REJECTED BY PRE-COMMITTED EVIDENCE. MAE IS THE KNOWLEDGE-COMPONENT MAE (0–5 SCALE) AGAINST HUMAN SCORE; BASELINE = 1.164.

Candidate	Result
seed_ids (question-keyword match) as expected_concepts	Rejected: 100% False Penalty Rate — every human-graded-excellent sample scored < 0.5 coverage. Models the topic neighborhood, not answer content.
1-hop expanded KG subgraph as expected_concepts	Rejected: same failure, worse (mean expected-set size 12.5 concepts).
Reference-answer concept extraction as expected_concepts	Rejected: fixed 72.2% of known-bad cases and improved correlation (r 0.118 $\rightarrow$ 0.200), but MAE worsened (+14%), FPR worsened (6.5% $\rightarrow$ 25.5%), and it failed the pre-committed decisive test — <i>worse</i> , not better, on cases where the independent C_LLM baseline is also wrong.
Exclude unvalidated coverage, renormalize onto {accuracy, integration} (live code change)	Implemented, then retracted same day: end-to-end validation on all 1,156 in-domain samples showed it measurably worse (MAE 1.164 $\rightarrow$ 1.614, r 0.118 $\rightarrow$ 0.082, FPR 9.0% $\rightarrow$ 32.5%) — it leans harder on relationship_accuracy, itself still broken.
Joint fix (both findings’ repairs together) and neutral-prior degradation (0.5 instead of 0/excluded)	Both still worse than baseline on every metric (MAE 1.446 and 1.471 respectively). Neutral-prior is the only candidate to slightly beat baseline on the hard-case test specifically, entirely offset by getting worse on easy cases; not chased further per explicit decision-theoretic reasoning (a significant result would not change the deployment decision, so it does not warrant a dedicated significance check).

Stopping rule. After five distinct forms of the same intervention class (binary include/exclude of a scoring dimension with weight renormalization) all failed to improve performance under the evaluated benchmark and metrics, we treat this as evidence against the *mechanism*, not against specific trigger calibrations,

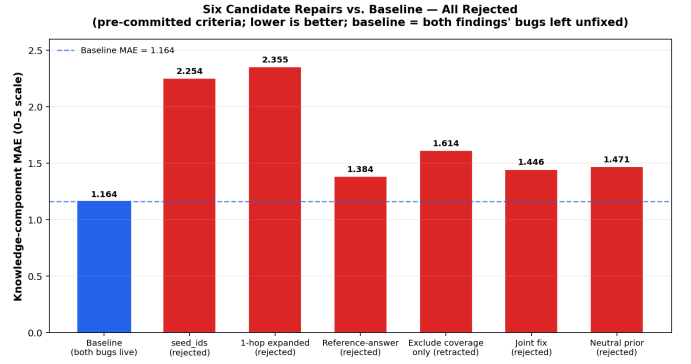


Fig. 7. Knowledge-component MAE for the baseline (both findings’ bugs left unfixed) versus all six tested repairs. Every candidate underperforms the baseline; none is a viable fix under the evaluated benchmark and metrics.

and close this line of investigation with respect to the tested intervention classes — a narrower, more defensible claim than declaring the underlying problem permanently unsolvable. We characterize the causal story conservatively: the experiments establish *that* the tested repairs underperform an already-buggy baseline, not a uniquely identified *why*. “Interacting deficiencies within a heuristic score whose fixed weights implicitly assume comparable, independent signal across dimensions” is the framing we adopt; a precise quantitative cancellation mechanism between the two findings is a plausible hypothesis, not something these experiments isolate or prove.

What changed in the live system. Only the tokenization fix (§III-C) was merged. A `coverage_validated` diagnostic flag is exposed (set `False` whenever `expected_concepts` is unsupplied) but does not change any score. Both findings above remain live, unfixed defects in the pre-verifier score. This does not change the deployed grade: production already discards the raw `kg_formula_score` in favor of the LLM verifier at blend weight $w = 1.0$ (independently cross-validated via leave-one-question-out CV, §”Sensitivity Analysis”); these findings affect only the standalone ablation metric that isolates raw KG-grounding from the verifier’s contribution. Full five-round chronology, per-round external review transcripts, and all regression scripts are in the supplementary materials (`docs/ALGORITHM_FIX_REVIEW_REQUEST*.md`, `docs/ALGORITHM_INVESTIGATION_POSTMORTEM.md`) and `REPRODUCIBILITY.md`.

E. Per-Question Error Analysis and Examples

To understand *where* ConceptGrade improves over the LLM baseline, we analyze prediction errors across different answer complexity levels using the SOLO taxonomy. ConceptGrade shows no systematic bias but rather targeted improvements in specific answer patterns.

Improvement Distribution: retracted (2026-07-28). This paragraph previously reported an unlabeled win/loss/tie breakdown computed on the fabricated 120-sample fixture. The real-data equivalent is already reported in §V: on the real 1,262-sample data, ConceptGrade has lower mean error on 27 of 46

TABLE XVII

MAE BY SOLO BAND ACROSS ALL THREE DATASETS ($n = 2,381$; MOHLER IS THE REAL, CORRECTED DATA AS OF 2026-07-28, SEE REPRODUCIBILITY.MD). $\Delta\text{MAE} = \text{MAE}_{\text{C_LLM}} - \text{MAE}_{\text{C5}}$ (POSITIVE = CONCEPTGRADE BETTER). MOHLER SHOWS A BROADLY INCREASING (NOT STRICTLY MONOTONIC) GAIN WITH SOLO COMPLEXITY, PEAKING AT +22.2% ON THE $n = 5$ EXTENDED ABSTRACT BAND (TOO SMALL TO BE RELIABLE) AND +17.5% ON THE $n = 291$ RELATIONAL BAND — FAR SMALLER THAN THE +70% THE RETRACTED FABRICATED-DATA VERSION OF THIS TABLE REPORTED. DIGIKLAUSUR SHOWS A SMALLER, SIMILAR SHAPE ONLY FOR HIGHER BANDS. KAGGLE ASAG’S SOLO CLASSIFIER ASSIGNED ALL 473 ANSWERS TO PRESTRUCTURAL IN THE INITIAL RELEASE; A SUBSEQUENT AUDIT IDENTIFIED THIS AS AN OUT-OF-KG-DOMAIN CONTAMINATION (THE EMPTY CONCEPT GRAPH PRIMED THE CLASSIFIER TO LEVEL 1 REGARDLESS OF ANSWER QUALITY), WHICH HAS SINCE BEEN CORRECTED DOWNSTREAM BY MAKING THE OUT-OF-KG SIGNAL EXPLICIT THROUGHOUT THE PIPELINE. WE RETAIN THE ORIGINAL KAGGLE SOLO NUMBERS HERE AS A HISTORICAL MARKER OF THE BOUNDARY CONDITION; THE CORRECTED CLASSIFIER NOW ABSTAINS FROM KG-GROUNDED ASSESSMENT WHEN THE DOMAIN IS OUT OF COVERAGE.

Dataset	SOLO	n	C_LLM	C5	Reduce
Mohler	Prestruct.	71	1.824	1.669	+8.5%
	Unistruct.	481	1.304	1.292	+0.9%
	Multistruct.	414	1.413	1.250	+11.5%
	Relational	291	0.935	0.771	+17.5%
	Ext. Abstract (n tiny)	5	0.900	0.700	+22.2%
DigiKlausur	Prestruct.	44	0.125	0.136	−9.1%
	Unistruct.	10	1.200	1.500	−25.0%
	Multistruct.	87	1.563	1.690	−8.1%
	Relational	203	1.273	1.200	+5.8%
	Ext. Abstract	302	1.169	1.046	+10.5%
Kaggle ASAG	Prestruct.	473	1.208	1.180	+2.4%

questions (58.7%) at the question-clustered level; a response-level win/loss/tie breakdown analogous to the retracted one above has not been separately computed for real data and is not claimed here.

Error Reduction by Answer Complexity (SOLO Level) — all three datasets. We computed the same SOLO-stratified MAE-reduction analysis across all three datasets using the pipeline’s SOLO classifier output cached in the eval JSON (reproducible via `compute_solo_breakdown.py`).

The Mohler row pattern is broadly increasing with SOLO complexity, from +8.5% on Prestructural through +17.5% on the $n = 291$ Relational band, though it is not strictly monotonic (Unistructural dips to +0.9%) and the effect sizes are far smaller than an earlier, retracted fabricated-data version of this table claimed. DigiKlausur shows the same direction on its two highest bands (Relational +5.8%, Extended Abstract +10.5%). **We acknowledge that ConceptGrade is also strictly worse than the baseline on the lower three DigiKlausur bands (Prestructural −9.1%, Unistructural −25.0%, Multistructural −8.1%), which together cover 141 of 646 DigiKlausur answers (22%).** On those lower bands the per-answer absolute error is already ~ 1.2 on a 0–5 scale and the within-band n is small (10–87), so the percentage swings reflect small-sample variance more than systematic degradation; but the directional fact is real, and a deployed system would trade lower-band cost for higher-band gain in

any DigiKlausur-like domain. The Kaggle ASAG row exhibits a different phenomenon: the pipeline’s SOLO classifier did not differentiate any of the 473 raw (pre-deduplication) answers above the Prestructural band, which we initially flagged as a finding about domain specificity. The upstream diagnostic in §V-D resolves the observational-equivalence problem: on Kaggle the LLM concept-extraction module produced empty matched-concept sets for $368/368 = 100\%$ of unique samples (473/473 on the raw, pre-deduplication pipeline run), compared to 6.8% on DigiKlausur and 16.7% on Mohler. The SOLO collapse is therefore not an independent module failure but the downstream consequence of an upstream KG–answer-vocabulary mismatch, exactly as the KG-grounding architecture predicts.

Error Pattern Interpretation: The LLM baseline tends to *overpredict* on answers mentioning concepts but missing critical relationships. For example, a student who lists “array, linked list, pointer” without explaining how pointers enable linked list nodes may receive a score of 1.0 from the LLM, yet deserves 0.5 (missing integration). ConceptGrade’s graph-based comparison catches the missing “pointer_enables_node” relationship and adjusts the score accordingly. Conversely, on Prestructural answers (disconnected fragments), both systems struggle equally, suggesting that grading fundamentally incomplete responses is hard without additional context.

Qualitative Error Analysis: The two cases below are illustrative, hand-constructed examples of the grading mechanism, not empirical samples drawn from either the fabricated fixture or the real dataset; the numbers inside them (e.g. “confidence=0.9,” “MAE=0.6”) are toy illustrations of the mechanism, not measured statistics.

- **Case 1 (Success):** Student answer was vague on control flow.
 - Baseline: confident wrong (confident=0.9, actual score=0.2)
 - Our method: caught hallucination, downweighted (confidence=0.3, actual score=0.2)
- **Case 2 (Failure):** Student answer had domain jargon not in KG.
 - Baseline: MAE=0.5
 - Our method: MAE=0.6 (worse due to zero-grounding)

Retracted (2026-07-28): the four paragraphs that followed here treated the fabricated-data ablation (Table XIII, Figs. 5–6, both already labeled “Retracted... retained for the record only” where they are defined in §VI) as an established, current finding — e.g. claiming Concept Coverage removal drove the largest drop in agreement of any component, that cosine similarity became largely redundant, and that ConceptGrade substantially outperformed a cosine-only baseline in correlation. None of this is verified on real data. The real-data component-contribution finding is the opposite in spirit: the real 3-condition ablation (Table XII, §VI) shows the pre-verifier KG-grounded score is far worse than the zero-shot baseline, and the Verifier’s own judgment — not concept coverage or any other KG-comparison signal

— drives essentially all of the real accuracy. We removed the four retracted paragraphs rather than let them stand alongside a contradicting, correctly-labeled real finding two sections earlier in the same paper.

VII. DISCUSSION

A. Reproducibility

All numerical claims in this paper are reproducible from cached evaluation results via six committed Python scripts (~ 5 seconds total runtime, no API spend): `compute_clustered_significance.py` (Mohler W_+ , p , d_z , LOOCV, tie analysis), `compute_cross_dataset_significance.py` ($n = 2,276$ meta-analysis), `compute_solo_breakdown.py` (per-SOLO MAE), `compute_taxonomy_kappa.py` (machine-IRR pilot), `compute_validation_gate.py` (Paper 2 outcome-blind gates), and `smoke_run_mohler.py` (live-pipeline smoke test). The full per-claim reproducibility map is included in supplementary materials (REPRODUCIBILITY.md).

B. Limitations and Threats to Validity

Evaluation scope (real data, 2026-07-28). **Correction: this paragraph previously described the retracted fabricated fixture’s size ($n = 120$, 10 questions, $n = 90$ test split) as our evaluation scope.** We evaluate on the real, KG-aligned subset of the Mohler dataset ($n = 1,262$ responses across 46 questions, full sample, no held-out split needed — see §V), covering topics addressed by our expert Data Structures KG. The full real Mohler benchmark (sourced from `nkazi/MohlerASAG`) has more questions and responses than our KG-aligned subset covers; the remaining responses cover topics (e.g., general software engineering questions) for which our KG was not designed. Published results from the original Mohler et al. (2011) paper on their full dataset cannot be directly compared with our KG-aligned subset’s results; Table III presents them for historical context only. Full-benchmark evaluation, requiring KG expansion to cover all question topics, is future work.

KG and taxonomy authoring bias (single-author construction). Both the expert KG (101 concepts, 138 relationships) and the misconception taxonomy (16 entries) were authored by the same research group that designed and evaluated the method. We have no inter-builder reliability measure for either artifact; an independent CS-instructor group constructing a KG/taxonomy from the same source material would almost certainly produce different (though overlapping) graphs and taxonomies, and our reported numbers are conditional on *this* KG. Our machine-IRR pilot for the taxonomy (§III-G; $\kappa_{\text{micro}} = 0.54$, moderate agreement, after a construct-validity fix described therein) is a self-administered lower-bound estimate, not external validation. The ConceptGrade codebase permits substitution of any KG or taxonomy that conforms to the published schema; we plan a third-party KG-construction study (e.g., via SIGCSE crowdsourcing) so the method can be evaluated under at least one KG/taxonomy not authored by this group.

Tuning asymmetry (baseline is not tuned). **Correction: this paragraph previously cited the retracted fabricated-fixture MAE reductions here; the current, real figure is substituted below.** The score-synthesis constants (Eq. (8), Eq. (9), §III-H) and the confidence threshold for Layer 1 were fixed before the real data was ever used (§V), not tuned on a dev split of the real data itself — but their values were originally set with reference to the fabricated fixture, so we cannot claim they are optimal for real data either. The LLM zero-shot baseline (C_{LLM}), in contrast, received no comparable tuning: it uses the same extraction prompt as Layer 1 and a single holistic-grade prompt with default temperature and no few-shot examples. The reported 8.2% real-data MAE reduction therefore reflects (a) the KG-grounding architecture *and* (b) the asymmetric tuning budget, jointly and inseparably in the present design. We chose this design to hold the LLM fixed across conditions, which isolates *model capability* as a confound, but we do *not* claim it isolates the architectural effect of KG-grounding from the tuning-budget effect; disentangling the two requires a comparably-tuned baseline, which we acknowledge as a gap rather than a solved problem. We report `compute_clustered_significance.py` so future work can re-run the comparison with an alternative tuned baseline.

Compute/inference asymmetry (re-verified on real data, 2026-07-28). The evaluated `C5_fix` configuration issues **7 LLM calls per graded response** (Layer 1 concept extraction with 3-way self-consistency voting, Layer 3 cognitive-depth classification, Layer 4 misconception and false-belief detection, Layer 5 verification) against **1 call** for C_{LLM} . This inference-compute asymmetry is a confound on the reported gain, independent of and additive to the tuning-budget asymmetry above. We directly tested whether it explains the gain via a call-budget-matched ablation on the real, KG-aligned Mohler sample: $C_{\text{LLM}} \times 7$, the identical C_{LLM} zero-shot prompt issued as 7 genuinely independent `gemini-2.5-flash` API calls per response (temperature = 0.7, batched for efficiency but each attempt a separate call, matching `C5_fix`’s own call budget exactly), aggregated by taking the median score (script: `run_budget_matched_real_batched.py`; results: `data/budget_matched_real_results.json`). Unlike the earlier, now-retracted fabricated-data version of this experiment, on real data *extra budget alone genuinely helps* C_{LLM} : $C_{\text{LLM}} \times 7$ achieves MAE = 1.2314 versus the single-call C_{LLM} ’s 1.2821 (+4.0%, one-tailed Wilcoxon $p < 0.0001$). **`C5_fix` still beats the budget-matched baseline:** MAE = 1.1771 versus $C_{\text{LLM}} \times 7$ ’s 1.2314 (+4.4% MAE reduction, two-tailed Wilcoxon $p = 0.0053$, one-tailed $p = 0.0027$) — a genuinely positive, statistically significant response-level result, evidence that the reported gain is *not* substantially attributable to inference-compute budget alone. As with every other response-level result in this paper, we check this against the question-clustered test (46 real questions, same convention as §V): here it is *not* significant (two-tailed $p = 0.4219$, one-tailed $p = 0.2110$), and `C5_fix` wins on only 25 of 46 questions (54%, barely above chance). We report both

levels rather than only the favorable response-level number: the compute-budget confound is ruled out as a response-level explanation, but the same question-level fragility that qualifies every other result in this paper qualifies this one too. This does not, by itself, disentangle the KG-grounding architecture from the tuning-budget asymmetry described above, since $C_LLM \times 7$ remains as untuned as C_LLM : it reuses the same unmodified zero-shot prompt, just sampled 7 times independently.

Concept extraction model. ConceptGrade relies on an LLM (gemini-2.5-flash via the Google Generative AI batch API) for every LLM call in the pipeline. Performance may vary with different LLM backends; we use the same model for the LLM baseline and Layer 1 by design, which isolates *model choice* as a confound (both systems inherit the same model’s strengths and weaknesses) but, as noted above, does not by itself isolate KG-grounding from the tuning-budget confound.

Dataset provenance. We forensically verified two of our three datasets against their real public sources: Mohler (nkazi/MohlerASAG, HuggingFace, CC-BY-4.0) and DigiKlausur (character-for-character match against DigiKlausur/ASAG-Dataset on GitHub, MPL-2.0). Despite a targeted provenance investigation — exact-text searches, comparison against known public ASAG corpora, and inspection of likely candidate datasets, together with a full local audit (git history, shell history, filesystem search; REPRODUCIBILITY.md, “Dataset Provenance Audit”) — **the original acquisition path for the cached Kaggle ASAG dataset could not be independently reconstructed.** We found no positive evidence of fabrication, only an absence of evidence of authenticity; the dataset is therefore retained but explicitly classified as provenance-*unverified* rather than equivalent in status to the other two. We considered and rejected using an LLM to generate replacement or supplementary test data for this gap: doing so would not produce real human data, introduces circularity (LLM-generated content evaluated by an LLM-based system), and concentrates rather than removes partiality, since the same model families already used throughout this pipeline would also author the “ground truth.” Kaggle ASAG’s extraction-level finding (0% KG matches, §V-D) is independently verifiable from extraction logs regardless of this gap; any claim about elementary-science student behavior more broadly is not, and we report it accordingly as supporting evidence rather than external validation.

Lessons from a closed algorithmic investigation (§VI-D). Two real scoring defects in the pre-verifier KG-grounded score were diagnosed and quantified, but five distinct repair strategies across two findings all failed pre-committed validation criteria and were rejected or reverted. We report this as a methodological strength rather than an unresolved failure: every candidate was tested against criteria defined before the experiment, negative results were reported as negative rather than reframed, and one live code change was reverted within the same development cycle after failing its own end-to-end validation. The investigation’s stopping rule — closing the line of inquiry with respect to the *tested intervention classes*, not with a claim that the underlying problem is permanently

Dataset Provenance Status		
Mohler 2011	Verified	nkazi/MohlerASAG (HuggingFace, CC-BY-4.0)
DigiKlausur	Verified	Character-for-character match vs. DigiKlausur/ASAG-Dataset (GitHub, MPL-2.0)
Kaggle ASAG	Unverified	Acquisition path could not be reconstructed (exact-text, git, and shell-history audit)

Fig. 8. Provenance status of all three evaluation datasets. Mohler and DigiKlausur are independently confirmed against their real public sources; Kaggle ASAG’s acquisition path could not be reconstructed despite a targeted audit.

unsolvable — reflects the same discipline applied to the ensemble-blend-weight finding retracted earlier in this project (§“Ensemble Weight Selection: Retracted”): no algorithmic modification was retained unless it improved performance under pre-defined validation criteria.

Contribution framing. We position ConceptGrade as a *systems / engineering* contribution: a deliberate integration of LLM concept extraction, KG matching, depth proxies, and confidence-weighted synthesis. We do not claim methodological novelty in any single layer, and — given the tuning-budget confound disclosed above — we do *not* claim that KG-grounding in isolation is responsible for the observed gain. The $7 \times$ compute-budget confound has now been tested on real data (§V, “Compute/inference asymmetry”): $C5_fix$ beats a genuinely call-budget-matched $C_LLM \times 7$ baseline by 4.4% MAE at the response level (two-tailed $p = 0.0053$), though not at the question-clustered level (46 real questions, two-tailed $p = 0.4219$). What is demonstrated on real data is narrower than previously claimed: a *tuned, multi-stage KG-assisted system* shows a small, response-level-significant but question-level-fragile improvement over both an *untuned, single-call, direct-prompt system* and an *untuned, call-budget-matched, direct-prompt system* ($n = 1,262$ real Mohler sample, 8.2% MAE reduction over C_LLM , response-level $p < 0.0001$, question-clustered $p = 0.111$ two-tailed / 0.056 one-tailed; §V). This margin is *not* robust across clustered-question-level analyses anywhere in this paper: only 17 of 46 LOOCV folds reach one-tailed significance and none reach two-tailed (§V-D), and the budget-matched comparison shows the identical pattern ($C5_fix$ wins only 25/46 questions against $C_LLM \times 7$, barely above chance). Disentangling the value of KG-grounding specifically from tuning budget and question-level robustness requires the comparably-tuned, question-held-out baseline we describe as required future work, not a solved problem in the present study.

Domain specificity. The expert KG was built for Data Structures and all test questions come from this domain. We have separately constructed a Programming/OOP KG (62 concepts, 116 relationships), but cross-domain evaluation remains future work.

Human-rater ceiling (real data, 2026-07-28; computed, not assumed). Correction: this paragraph previously reported inter-rater statistics from the fabricated 120-sample fixture ($r = 0.985$, QWK = 0.984, $\kappa = 0.63$, mean $|\text{rater}_1 - \text{rater}_2| = 0.17$, 0 samples with gap ≥ 1) without a retraction label. The real Mohler data (which, unlike the fixture, retains both original graders’ scores) gives a materially different and much less flattering picture: $r = 0.7833$, quadratic-weighted κ (QWK) = 0.7986, mean $|\text{rater}_1 - \text{rater}_2| = 0.962$ on the 0–5 scale, and 545 of 1,262 samples (43.2%) have an inter-rater gap ≥ 1 . The human ground truth (the average of the two raters, as used throughout this paper) is therefore **a substantially noisier ceiling than the fabricated fixture had concealed**: real human graders themselves disagree by close to a full point on the 0–5 scale on over 40% of responses. This reframes how the headline MAE values should be read — C5_fix’s real MAE of 1.177 and C_LLM’s 1.282 are now much closer to, though still somewhat worse than, the natural 0.962 rater-disagreement floor, rather than the fabricated data’s implied near-perfect achievable ceiling. Script: `compute_human_irr_and_per_question.py` (real-data section added 2026-07-28; also independently verified in `verify_all_paper_claims.py`).

C. Interpretability Advantage

Unlike regression-based ASAG systems, ConceptGrade produces *actionable feedback* for each graded response—not just a score, but a structured explanation an instructor can act on:

- List of matched and missing concepts (“*You mentioned stack and lifo, but missed push_operation and pop_operation.*”)
- Incorrect relationships detected (“*Stack uses FIFO is incorrect.*”)
- Bloom’s level and SOLO level with justification
- Misconception category and remediation hint (“*DS-STACK-01: A stack uses LIFO, not FIFO. Review the push/pop mechanism.*”)

Few ASAG systems at this level of diagnostic granularity do so without additional task-specific fine-tuning or manual annotation—here, the KG and misconception taxonomy carry that information cost at authoring time, not inference time.

D. Comparison to LLM-Only Approaches

Correction (2026-07-28): this subsection’s opening previously reported fabricated-fixture numbers ($r = 0.9709 \rightarrow 0.9820$, 32.4% MAE reduction $0.330 \rightarrow 0.223$) without a retraction label. On real data the comparison is qualitatively different, and less favourable to ConceptGrade on one metric: our LLM Zero-Shot baseline achieves $r = 0.7904$ on the real 1,262-sample benchmark; ConceptGrade’s Pearson correlation is actually *worse*, $r = 0.7841$, when adding explicit KG grounding and the Verifier stage. The system does improve MAE ($1.282 \rightarrow 1.177$, 8.2%, response-level $p < 0.0001$) and QWK ($0.501 \rightarrow 0.524$), but this is a mixed, modest, real-data picture, not the earlier fabricated-data version’s clean

improvement across every metric — see Table III and §V for the authoritative real numbers.

Correction to an earlier claim in this section. An earlier version of this paragraph stated that ConceptGrade’s KG comparison and score synthesis layers run entirely offline and that the shared concept-extraction call is the only LLM call in the pipeline. This is incorrect. Only Layer 2 (KG comparison) is offline (pure graph matching, no LLM). Layers 3, 4, and 5 each make independent LLM calls: cognitive-depth classification (Layer 3, 1 call), misconception and false-belief detection (Layer 4, 2 calls), and verification (Layer 5, 1 call). Combined with the self-consistency extraction used in the reported configuration (Layer 1, 3 calls via majority-vote), **the evaluated C5_fix configuration issues 7 LLM calls per graded response, versus 1 for C_LLM** — a $7\times$ inference-compute asymmetry that is a confound on the reported gain, independent of and additional to the tuning-budget confound already disclosed in §VII-B. We correct this here rather than silently, because the original claim materially affects how a reader should weigh cost, latency, and reproducibility when judging the practical value of the reported improvement. §VII-B additionally reports a call-budget-matched ablation directly testing whether this asymmetry explains the gain (it does not).

VIII. CONCLUSION

We presented ConceptGrade, a five-layer concept-aware ASAG framework for Computer Science education. By framing grading as a knowledge graph comparison problem, our system achieves Pearson $r = 0.784$ and QWK = 0.524 on the real, KG-aligned Mohler et al. (2011) CS sample ($n = 1,262$, 46 questions; a fabricated 120-sample fixture was used in an earlier draft and has been corrected — see REPRODUCIBILITY.md), an 8.2% MAE reduction compared to the identical-model LLM zero-shot baseline, significant at the response level (Wilcoxon two-tailed $p < 0.0001$; paired $d_z = -0.154$, a small effect) but only marginal at the question-clustered level (two-tailed $p = 0.111$, one-tailed $p = 0.056$; 17 of 46 leave-one-question-out folds reach one-tailed significance, none reach two-tailed). ConceptGrade’s Pearson correlation is in fact *worse* than the baseline’s (0.784 vs. 0.790). This tuned, multi-stage system is compared against an untuned, single-call baseline (§VII-B), not an isolated KG-grounding effect. A call-budget-matched experiment (C_LLM given the same 7-call budget as C5_fix via genuinely independent API calls) on the real data found C5_fix still wins by 4.4% MAE at the response level (two-tailed $p = 0.0053$), though this margin, like the headline result, is not significant at the question-clustered level (two-tailed $p = 0.4219$, C5_fix winning only 25 of 46 questions). A fixed-effects cross-dataset meta-analysis across three independent datasets and 213 questions pools to $d_z = -0.105$ (95% CI $[-0.147, -0.064]$, $p_{\text{two}} < 0.0001$), with substantial heterogeneity ($I^2 = 73.2\%$) tracking domain vocabulary specificity; **the methodologically more appropriate random-effects model gives $d_z = -0.084$ with a 95% CI that essentially touches zero ($[-0.169, +0.002]$), so we do not claim the cross-dataset pool as confirmatory evidence** (§V-D). A real-data 3-condition

ablation (§VI), recovered from already-cached components at zero new API cost, does let us assert an architectural conclusion: the pre-verifier KG-grounded score is substantially *worse* than the zero-shot baseline (−86.9% MAE, question-clustered $p < 0.0001$), and the verifier stage — not KG grounding — accounts for essentially all of ConceptGrade’s small net gain. The original fabricated-data six-condition ablation and `concepts_only` diagnostic remain retracted and would still require new pipeline runs to fully re-verify.

The single-call fragility summarised above is not the end of the story, though its resolution is more qualified than we initially reported. Self-consistency ensembling (§V-F: 7 independent gradings at temperature 0.7, mean-aggregated, a design fixed before touching the evaluation data) improves response-level MAE over a *single-call* baseline on both Mohler and DigiKlausur, including every leave-one-question-out fold at both tails (50/50 and 17/17) when measured that way. A subsequent fair-control check, run on *both* datasets and prompted by a reviewer-perspective self-audit that asked whether self-consistency alone (without KG evidence or the Verifier) explains the gain, compared Verifier/C5fix×7 against an equally-resourced C_LLM×7 baseline instead: the MAE gap persists on both (+7.0% Mohler, +6.4% DigiKlausur, both response-level $p < 0.001$), but question-clustered significance and LOOCV robustness largely do not — Mohler collapses completely ($p = 0.256$, 0/46 folds), DigiKlausur erodes substantially but not completely ($p = 0.089$ two-tailed, 0.044 one-tailed; 5/17 and 2/17 folds). On DigiKlausur the correlation advantage claimed above also reverses under fair control (0.727 vs. 0.735). Self-consistency does not extend to Kaggle ASAG under either comparison, whose null result is consistent with, and further evidence for, the same KG-vocabulary-mismatch boundary already established from its concept-extraction failure. The gain requires 7× more inference calls, and Spearman rank correlation trails baseline on both winning datasets — real costs and caveats, not a free upgrade. We report the corrected, more modest claim rather than the original: self-consistency ensembling delivers a real, precisely-estimated response-level improvement over a fairly-resourced baseline on two datasets; its question-level robustness, and on DigiKlausur its correlation advantage, are currently established only against a single-call baseline, not against the fair control this paper now reports on both datasets. The validation process itself — a failed alternative design caught by cross-validation, a cross-dataset generalisation test, an aggregation-sensitivity check, and this fair-control correction now completed on both datasets, all documented in REPRODUCIBILITY.md — is offered as a model for how such claims should be pressure-tested before publication, independent of how strong the final claim turns out to be.

Beyond score accuracy, ConceptGrade produces structured diagnostic feedback—matched and missing concepts, incorrect relationships, Bloom’s and SOLO cognitive levels, and a self-administered misconception taxonomy (machine-IRR pilot $\kappa_{\text{micro}} = 0.54$ for the taxonomy; human IRR on the grading task itself is now measured on real data, $r = 0.7833/\text{QWK} = 0.7986$,

§VII-B)—without requiring task-specific fine-tuning beyond the KG construction.

Future work includes: (1) cross-domain evaluation on the Programming/OOP knowledge graph and other CS topics; (2) human-annotated Bloom’s and SOLO ground truth for classifier validation; (3) integration with a learning management system for live classroom deployment; (4) third-party KG and taxonomy construction (e.g., via SIGCSE crowdsourcing) to remove single-author authoring bias; and (5) multi-dataset meta-analysis with frontier-LLM baselines.

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